Generally speaking, suppose nonzero vectors **A**, **B**, **C** have the same initial point and are not coplanar. Then **A**, **B**, **C** are said to form a *right-handed system* or *dextral system* if a right-threaded screw rotated through an angle less than 180° from **A** to **B** will advance in the direction **C** as shown in Fig. 1-4(b).

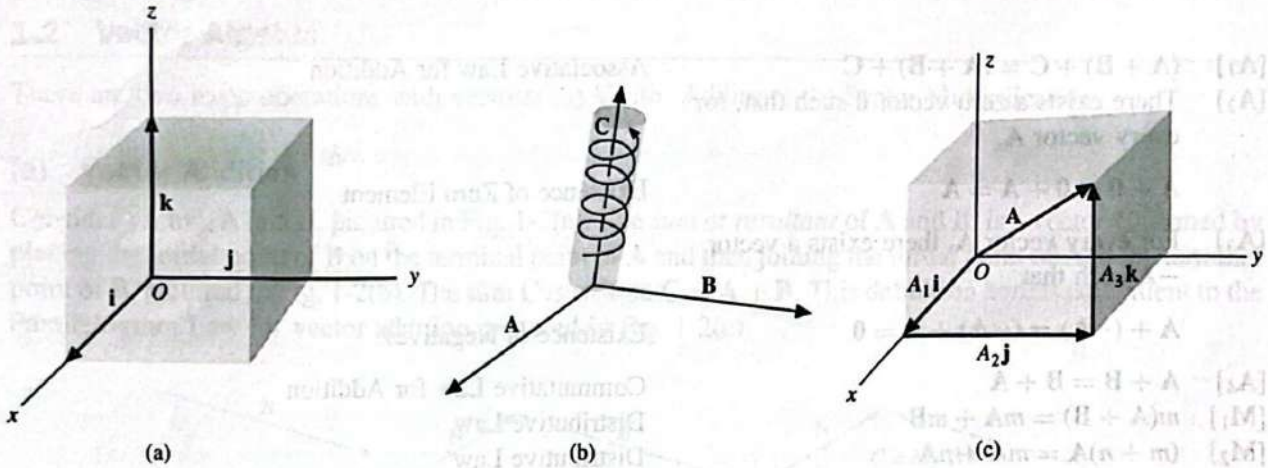


Fig. 1-4

Components of a Vector

Any vector **A** in three dimensions can be represented with an initial point at the origin $O = (0, 0, 0)$ and its end point at some point, say, (A_1, A_2, A_3). Then the vectors $A_1\mathbf{i}$, $A_2\mathbf{j}$, $A_3\mathbf{k}$ are called the *component vectors* of **A** in the x, y, z directions, and the scalars A_1, A_2, A_3 are called the *components* of **A** in the x, y, z directions, respectively. (See Fig. 1-4(c).)

The sum of $A_1\mathbf{i}$, $A_2\mathbf{j}$, and $A_3\mathbf{k}$ is the vector **A**, so we may write

$$\mathbf{A} = A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}$$

The magnitude of **A** follows:

$$|\mathbf{A}| = \sqrt{A_1^2 + A_2^2 + A_3^2}$$

Consider a point $P(x, y, z)$ in space. The vector **r** from the origin O to the point P is called the *position* vector (or *radius* vector). Thus **r** may be written

$$\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$$

It has magnitude $|\mathbf{r}| = \sqrt{x^2 + y^2 + z^2}$.

The following proposition applies.

PROPOSITION 1.3: Suppose $\mathbf{A} = A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}$ and $\mathbf{B} = B_1\mathbf{i} + B_2\mathbf{j} + B_3\mathbf{k}$. Then

(i) $\mathbf{A} + \mathbf{B} = (A_1 + B_1)\mathbf{i} + (A_2 + B_2)\mathbf{j} + (A_3 + B_3)\mathbf{k}$

(ii) $m\mathbf{A} = m(A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}) = (mA_1)\mathbf{i} + (mA_2)\mathbf{j} + (mA_3)\mathbf{k}$

EXAMPLE 1.2 Suppose $\mathbf{A} = 3\mathbf{i} + 5\mathbf{j} - 2\mathbf{k}$ and $\mathbf{B} = 4\mathbf{i} - 8\mathbf{j} + 7\mathbf{k}$.

(a) To find $\mathbf{A} + \mathbf{B}$, add corresponding components, obtaining $\mathbf{A} + \mathbf{B} = 7\mathbf{i} - 3\mathbf{j} + 5\mathbf{k}$

(b) To find $3\mathbf{A} - 2\mathbf{B}$, first multiply by the scalars and then add:

$$3\mathbf{A} - 2\mathbf{B} = (9\mathbf{i} + 15\mathbf{j} - 6\mathbf{k}) + (-8\mathbf{i} + 16\mathbf{j} - 14\mathbf{k}) = \mathbf{i} + 31\mathbf{j} - 20\mathbf{k}$$

(c) To find $|\mathbf{A}|$ and $|\mathbf{B}|$, take the square root of the sum of the squares of the components:

$$|\mathbf{A}| = \sqrt{9 + 25 + 4} = \sqrt{38} \quad \text{and} \quad |\mathbf{B}| = \sqrt{16 + 64 + 49} = \sqrt{129}$$

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(b) *Graphically*, the resultant of $\mathbf{r}_1$ and $\mathbf{r}_2$ is obtained as the diagonal **OR** of parallelogram *OPRQ*. *Analytically*, the resultant of $\mathbf{r}_1$ and $\mathbf{r}_2$ is given by

$$\mathbf{r}_1 + \mathbf{r}_2 = (2\mathbf{i} + 4\mathbf{j} + 3\mathbf{k}) + (\mathbf{i} - 5\mathbf{j} + 2\mathbf{k}) = 3\mathbf{i} - \mathbf{j} + 5\mathbf{k}$$

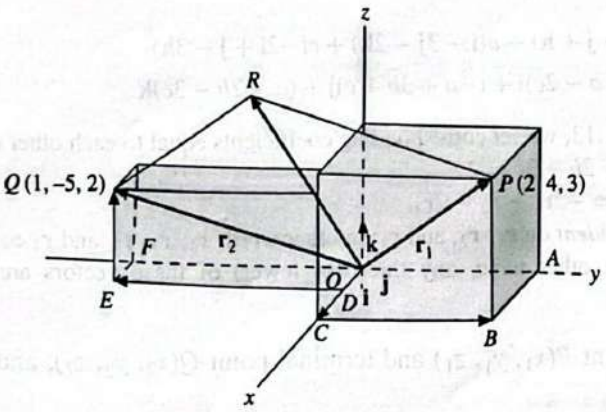


Fig. 1-18

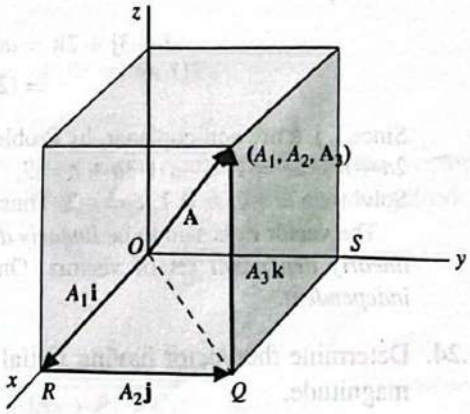


Fig. 1-19

1.20. Prove that the magnitude of the vector $\mathbf{A} = A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}$, pictured in Fig. 1-19, is $|\mathbf{A}| = \sqrt{A_1^2 + A_2^2 + A_3^2}$.

Solution

By the Pythagorean theorem,

$$(\overline{OP})^2 = (\overline{OQ})^2 + (\overline{QP})^2$$

where $\overline{OP}$ denotes the magnitude of vector **OP**, and so on. Similarly, $(\overline{OQ})^2 = (\overline{OR})^2 + (\overline{RQ})^2$.

Then $(\overline{OP})^2 = (\overline{OR})^2 + (\overline{RQ})^2 + (\overline{QP})^2$ or $A^2 = A_1^2 + A_2^2 + A_3^2$ (i.e., $\mathbf{A} = \sqrt{A_1^2 + A_2^2 + A_3^2}$).

1.21. Given the radius vectors $\mathbf{r}_1 = 3\mathbf{i} - 2\mathbf{j} + \mathbf{k}$, $\mathbf{r}_2 = 3\mathbf{i} + 4\mathbf{j} + 9\mathbf{k}$, $\mathbf{r}_3 = -\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$. Find the magnitudes of: (a) $\mathbf{r}_3$, (b) $\mathbf{r}_1 + \mathbf{r}_2 + \mathbf{r}_3$, (c) $\mathbf{r}_1 - \mathbf{r}_2 + 4\mathbf{r}_3$.

Solution

(a) $|\mathbf{r}_3| = |-\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}| = \sqrt{(-1)^2 + (2)^2 + (2)^2} = 3$.

(b) $\mathbf{r}_1 + \mathbf{r}_2 + \mathbf{r}_3 = 3\mathbf{i} + 4\mathbf{j} + 12\mathbf{k}$, hence $|\mathbf{r}_1 + \mathbf{r}_2 + \mathbf{r}_3| = \sqrt{9 + 16 + 144} = \sqrt{169} = 13$.

(c) $\mathbf{r}_1 - \mathbf{r}_2 + 4\mathbf{r}_3 = 2\mathbf{i} + 2\mathbf{j} = \sqrt{4+4} = \sqrt{8} = 2\sqrt{2}$.

1.22. Find a unit vector $\mathbf{u}$ parallel to the resultant $\mathbf{R}$ of vectors $\mathbf{r}_1 = 2\mathbf{i} + 4\mathbf{j} - 5\mathbf{k}$ and $\mathbf{r}_2 = -\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$.

Solution

Resultant $\mathbf{R} = \mathbf{r}_1 + \mathbf{r}_2 = (2\mathbf{i} + 4\mathbf{j} - 5\mathbf{k}) + (-\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}) = \mathbf{i} + 2\mathbf{j} - 2\mathbf{k}$. Also,

$$\text{Magnitude of } \mathbf{R} = |\mathbf{R}| = |\mathbf{i} + 2\mathbf{j} - 2\mathbf{k}| = \sqrt{(1)^2 + (2)^2 + (-2)^2} = 3.$$

Then $\mathbf{u}$ is equal to $\mathbf{R}/|\mathbf{R}|$. That is,

$$\mathbf{u} = \mathbf{R}/|\mathbf{R}| = (\mathbf{i} + 2\mathbf{j} - 2\mathbf{k})/3 = (1/3)\mathbf{i} + (2/3)\mathbf{j} - (2/3)\mathbf{k}$$

Check: $|(1/3)\mathbf{i} + (2/3)\mathbf{j} - (2/3)\mathbf{k}| = \sqrt{(1/3)^2 + (2/3)^2 + (-2/3)^2} = 1$.

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1.23. Suppose $\mathbf{r}_1 = 2\mathbf{i} - \mathbf{j} + \mathbf{k}$, $\mathbf{r}_2 = \mathbf{i} - 3\mathbf{j} - 2\mathbf{k}$, $\mathbf{r}_3 = -2\mathbf{i} + \mathbf{j} - 3\mathbf{k}$. Write $\mathbf{r}_4 = \mathbf{i} + 3\mathbf{j} + 2\mathbf{k}$ as a linear combination of $\mathbf{r}_1$, $\mathbf{r}_2$, $\mathbf{r}_3$; that is, find scalars a, b, c such that $\mathbf{r}_4 = a\mathbf{r}_1 + b\mathbf{r}_2 + c\mathbf{r}_3$.

Solution

We require

$$\begin{aligned}\mathbf{i} + 3\mathbf{j} + 2\mathbf{k} &= a(2\mathbf{i} - \mathbf{j} + \mathbf{k}) + b(\mathbf{i} - 3\mathbf{j} - 2\mathbf{k}) + c(-2\mathbf{i} + \mathbf{j} - 3\mathbf{k}) \\ &= (2a + b - 2c)\mathbf{i} + (-a + 3b + c)\mathbf{j} + (a - 2b - 3c)\mathbf{k}\end{aligned}$$

Since $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$ are non-coplanar, by Problem 1.13, we set corresponding coefficients equal to each other obtaining
$2a + b - 2c = 1$, $-a + 3b + c = 3$, $a - 2b - 3c = 2$
Solving, $a = -2$, $b = 1$, $c = -2$. Thus $\mathbf{r}_4 = -2\mathbf{r}_1 + \mathbf{r}_2 - 2\mathbf{r}_3$.

The vector $\mathbf{r}_4$ is said to be *linearly dependent* on $\mathbf{r}_1$, $\mathbf{r}_2$, and $\mathbf{r}_3$; in other words $\mathbf{r}_1$, $\mathbf{r}_2$, $\mathbf{r}_3$, and $\mathbf{r}_4$ constitute a *linearly dependent* set of vectors. On the other hand, any three (or fewer) of these vectors are *linearly independent*.

1.24. Determine the vector having initial point $P(x_1, y_1, z_1)$ and terminal point $Q(x_2, y_2, z_2)$, and find its magnitude.

Solution

Consider Fig. 1-20. The position vectors of P and Q are, respectively,

$$\mathbf{r}_1 = x_1\mathbf{i} + y_1\mathbf{j} + z_1\mathbf{k} \quad \text{and} \quad \mathbf{r}_2 = x_2\mathbf{i} + y_2\mathbf{j} + z_2\mathbf{k}$$

Then $\mathbf{r}_1 + \mathbf{PQ} = \mathbf{r}_2$ or

$$\begin{aligned}\mathbf{PQ} = \mathbf{r}_2 - \mathbf{r}_1 &= (x_2\mathbf{i} + y_2\mathbf{j} + z_2\mathbf{k}) - (x_1\mathbf{i} + y_1\mathbf{j} + z_1\mathbf{k}) \\ &= (x_2 - x_1)\mathbf{i} + (y_2 - y_1)\mathbf{j} + (z_2 - z_1)\mathbf{k}.\end{aligned}$$

Magnitude of $\mathbf{PQ} = \overline{PQ} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$. Note that this is the distance between points P and Q.

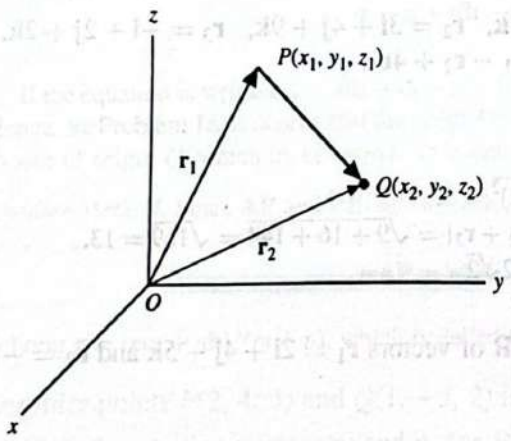


Fig. 1-20

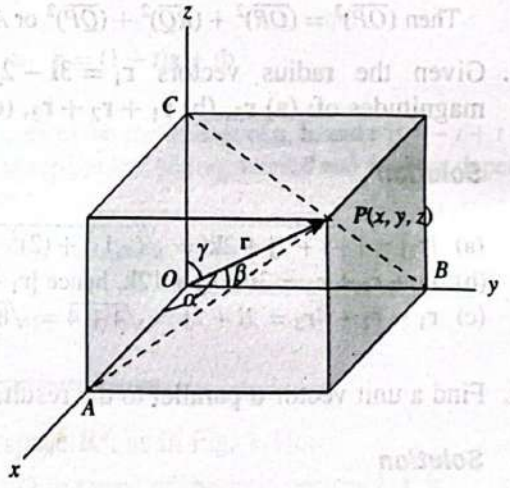


Fig. 1-21

1.25. Determine the angles α, β, and γ that the vector $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ makes with the positive directions of the coordinate axes and show that

$$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1.$$

Solution

Referring to Fig. 1-21, triangle OAP is a right triangle with right angle at A; then $\cos\alpha = x/|\mathbf{r}|$. Similarly, from right triangles OBP and OCP, $\cos\beta = y/|\mathbf{r}|$ and $\cos\gamma = z/|\mathbf{r}|$, respectively. Also, $|\mathbf{r}| = r = \sqrt{x^2 + y^2 + z^2}$.

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1.46. Prove that the line joining the midpoints of two sides of a triangle is parallel to the third side and has one half of its magnitude.

1.47. Prove that the medians of a triangle meet in a common point, which is a point of trisection of the medians.

1.48. Prove that the angle bisectors of a triangle meet in a common point.

1.49. Let the position vectors of points P and Q relative to the origin O is given by vectors $\mathbf{p}$ and $\mathbf{q}$, respectively. Suppose R is a point which divides PQ into segments that are in the ratio $m:n$. Show that the position vector of R is given by $\mathbf{r} = (m\mathbf{p} + n\mathbf{q})/(m+n)$ and that this is independent of the origin.

1.50. A quadrilateral $ABCD$ has masses of 1, 2, 3, and 4 units located, respectively, at its vertices $A(-1, -2, 2)$, $B(3, 2, -1)$, $C(1, -2, 4)$, and $D(3, 1, 2)$. Find the coordinates of the centroid.

1.51. Show that the equation of a plane which passes through three given points A, B, and C not in the same straight line and having position vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ relative to an origin O, can be written

$$\mathbf{r} = \frac{m\mathbf{a} + n\mathbf{b} + p\mathbf{c}}{m+n+p}$$

where m, n, p are scalars. Verify that the equation is independent of the origin.

1.52. The position vectors of points P and Q are given by $\mathbf{r}_1 = 2\mathbf{i} + 3\mathbf{j} - \mathbf{k}$, $\mathbf{r}_2 = 4\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$. Determine $\mathbf{PQ}$ in terms of $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$, and find its magnitude.

1.53. Suppose $\mathbf{A} = 3\mathbf{i} - \mathbf{j} - 4\mathbf{k}$, $\mathbf{B} = -2\mathbf{i} + 4\mathbf{j} - 3\mathbf{k}$, $\mathbf{C} = \mathbf{i} + 2\mathbf{j} - \mathbf{k}$. Find
(a) $2\mathbf{A} - \mathbf{B} + 3\mathbf{C}$, (b) $|\mathbf{A} + \mathbf{B} + \mathbf{C}|$, (c) $|3\mathbf{A} - 2\mathbf{B} + 4\mathbf{C}|$, (d) a unit vector parallel to $3\mathbf{A} - 2\mathbf{B} + 4\mathbf{C}$.

1.54. The following forces act on a particle P: $\mathbf{F}_1 = 2\mathbf{i} + 3\mathbf{j} - 5\mathbf{k}$, $\mathbf{F}_2 = -5\mathbf{i} + \mathbf{j} + 3\mathbf{k}$, $\mathbf{F}_3 = \mathbf{i} - 2\mathbf{j} + 4\mathbf{k}$, $\mathbf{F}_4 = 4\mathbf{i} - 3\mathbf{j} - 2\mathbf{k}$, measured in pounds. Find (a) the resultant of the forces, (b) the magnitude of the resultant.

1.55. In each case, determine whether the vectors are linearly independent or linearly dependent:
(a) $\mathbf{A} = 2\mathbf{i} + \mathbf{j} - 3\mathbf{k}$, $\mathbf{B} = \mathbf{i} - 4\mathbf{k}$, $\mathbf{C} = 4\mathbf{i} + 3\mathbf{j} - \mathbf{k}$, (b) $\mathbf{A} = \mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$, $\mathbf{B} = 2\mathbf{i} - 4\mathbf{j} - \mathbf{k}$, $\mathbf{C} = 3\mathbf{i} + 2\mathbf{j} - \mathbf{k}$.

1.56. Prove that any four vectors in three dimensions must be linearly dependent.

1.57. Show that a necessary and sufficient condition that the vectors $\mathbf{A} = A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}$, $\mathbf{B} = B_1\mathbf{i} + B_2\mathbf{j} + B_3\mathbf{k}$, $\mathbf{C} = C_1\mathbf{i} + C_2\mathbf{j} + C_3\mathbf{k}$ be linearly independent is that the determinant

$$\begin{vmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{vmatrix}$$ be different from zero.

1.58. (a) Prove that the vectors $\mathbf{A} = 3\mathbf{i} + \mathbf{j} - 2\mathbf{k}$, $\mathbf{B} = -\mathbf{i} + 3\mathbf{j} + 4\mathbf{k}$, $\mathbf{C} = 4\mathbf{i} - 2\mathbf{j} - 6\mathbf{k}$ can form the sides of a triangle.
(b) Find the lengths of the medians of the triangle.

1.59. Given the scalar field defined by $\phi(x, y, z) = 4yx^3 + 3xyz - z^2 + 2$. Find (a) $\phi(1, -1, -2)$, (b) $\phi(0, -3, 1)$.

ANSWERS TO SUPPLEMENTARY PROBLEMS

1.31. (a) s, (b) v, (c) s, (d) s, (e) v, (f) s, (g) s, (h) v, (i) s.

1.32. Magnitude 304.1 ($50\sqrt{37}$), direction 25°17′ north of east (arcsin $3\sqrt{111}/74$).

1.33. Magnitude: 20.9 mi, direction 21°39′ south of west.

1.34. 3AD.

1.37. Straight line course upstream making an angle 34°28′ with the shore line. 1 hr 25 min.

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CHAPTER 2

The DOT and CROSS Product

2.1 Introduction

Operations of vector addition and scalar multiplication were defined for our vectors and scalars in Chapter 1. Here, we define two new operations of multiplication for our vectors. One of the operations, the DOT product, yields a scalar, while the other operation, the CROSS product yields a vector. We then combine these operations to define certain triple products.

2.2 Dot or Scalar Product

The dot or scalar product of two vectors **A** and **B**, denoted by $\mathbf{A} \cdot \mathbf{B}$ (read: **A** dot **B**), is defined as the product of the magnitudes of **A** and **B** and the cosine of the angle θ between them. In symbols,

$$\mathbf{A} \cdot \mathbf{B} = |\mathbf{A}||\mathbf{B}| \cos \theta, \qquad 0 \leq \theta \leq \pi$$

We emphasize that $\mathbf{A} \cdot \mathbf{B}$ is a scalar and not a vector.

The following proposition applies.

PROPOSITION 2.1: Suppose **A**, **B**, and **C** are vectors and m is a scalar. Then the following laws hold:

(i) $\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A}$ Commutative Law for Dot Products
(ii) $\mathbf{A} \cdot (\mathbf{B} + \mathbf{C}) = \mathbf{A} \cdot \mathbf{B} + \mathbf{A} \cdot \mathbf{C}$ Distributive Law
(iii) $m(\mathbf{A} \cdot \mathbf{B}) = (m\mathbf{A}) \cdot \mathbf{B} = \mathbf{A} \cdot (m\mathbf{B}) = (\mathbf{A} \cdot \mathbf{B})m$
(iv) $\mathbf{i} \cdot \mathbf{i} = \mathbf{j} \cdot \mathbf{j} = \mathbf{k} \cdot \mathbf{k} = 1, \qquad \mathbf{i} \cdot \mathbf{j} = \mathbf{j} \cdot \mathbf{k} = \mathbf{k} \cdot \mathbf{i} = 0$
(v) If $\mathbf{A} \cdot \mathbf{B} = 0$ and **A** and **B** are not null vectors, then **A** and **B** are perpendicular.

There is a simple formula for $\mathbf{A} \cdot \mathbf{B}$ when the unit vectors **i**, **j**, **k** are used.

PROPOSITION 2.2: Given $\mathbf{A} = A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}$ and $\mathbf{B} = B_1\mathbf{i} + B_2\mathbf{j} + B_3\mathbf{k}$. Then

$$\mathbf{A} \cdot \mathbf{B} = A_1B_1 + A_2B_2 + A_3B_3$$

COROLLARY 2.3: Suppose $\mathbf{A} = A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}$. Then $\mathbf{A} \cdot \mathbf{A} = A_1^2 + A_2^2 + A_3^2$.

EXAMPLE 2.1 Given $\mathbf{A} = 4\mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$, $\mathbf{B} = 5\mathbf{i} - \mathbf{j} - 2\mathbf{k}$, $\mathbf{C} = 3\mathbf{i} + \mathbf{j} + 7\mathbf{k}$. Then:

$$\mathbf{A} \cdot \mathbf{B} = (4)(5) + (2)(-1) + (-3)(-2) = 20 - 2 + 6 = 24, \quad \mathbf{A} \cdot \mathbf{C} = 12 + 2 - 21 = -7,$$

$$\mathbf{B} \cdot \mathbf{C} = 15 - 1 - 14 = 0, \quad \mathbf{A} \cdot \mathbf{A} = 4^2 + 2^2 + (-3)^2 = 16 + 4 + 9 = 29$$

Thus vectors **B** and **C** are perpendicular.

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2.3 Cross Product

The cross product of vectors **A** and **B** is a vector $\mathbf{C} = \mathbf{A} \times \mathbf{B}$ (read: **A** cross **B**) defined as follows. The magnitude of $\mathbf{C} = \mathbf{A} \times \mathbf{B}$ is equal to the product of the magnitudes of **A** and **B** and the sine of the angle θ between them. The direction of $\mathbf{C} = \mathbf{A} \times \mathbf{B}$ is perpendicular to the plane of **A** and **B** so that **A**, **B**, and **C** form a right-handed system. In symbols,

$$\mathbf{A} \times \mathbf{B} = |\mathbf{A}||\mathbf{B}| \sin\theta\, \mathbf{u} \qquad 0 \le \theta \le \pi$$

where **u** is a unit vector indicating the direction of $\mathbf{A} \times \mathbf{B}$. [Thus **A**, **B**, and **u** form a right-handed system.] If $\mathbf{A} = \mathbf{B}$, or if **A** is parallel to **B**, then $\sin\theta = 0$ and we define $\mathbf{A} \times \mathbf{B} = \mathbf{0}$.

The following proposition applies.

PROPOSITION 2.4: Suppose **A**, **B**, and **C** are vectors and m is a scalar. Then the following laws hold:

(i) $\mathbf{A} \times \mathbf{B} = -(\mathbf{B} \times \mathbf{A})$ Commutative Law for Cross Products Fails
(ii) $\mathbf{A} \times (\mathbf{B} + \mathbf{C}) = \mathbf{A} \times \mathbf{B} + \mathbf{A} \times \mathbf{C}$ Distributive Law
(iii) $m(\mathbf{A} \times \mathbf{B}) = (m\mathbf{A}) \times \mathbf{B} = \mathbf{A} \times (m\mathbf{B}) = (\mathbf{A} \times \mathbf{B})m$
(iv) $\mathbf{i} \times \mathbf{i} = \mathbf{j} \times \mathbf{j} = \mathbf{k} \times \mathbf{k} = \mathbf{0}$, $\mathbf{i} \times \mathbf{j} = \mathbf{k}$, $\mathbf{j} \times \mathbf{k} = \mathbf{i}$, $\mathbf{k} \times \mathbf{i} = \mathbf{j}$
(v) If $\mathbf{A} \times \mathbf{B} = \mathbf{0}$ and **A** and **B** are not null vectors, then **A** and **B** are parallel.
(vi) The magnitude of $\mathbf{A} \times \mathbf{B}$ is the same as the area of a parallelogram with sides **A** and **B**.

There is a simple formula for $\mathbf{A} \times \mathbf{B}$ when the unit vectors **i**, **j**, **k** are used.

PROPOSITION 2.5: Given $\mathbf{A} = A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}$ and $\mathbf{B} = B_1\mathbf{i} + B_2\mathbf{j} + B_3\mathbf{k}$. Then

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \end{vmatrix} = \begin{vmatrix} A_2 & A_3 \\ B_2 & B_3 \end{vmatrix}\mathbf{i} - \begin{vmatrix} A_1 & A_3 \\ B_1 & B_3 \end{vmatrix}\mathbf{j} + \begin{vmatrix} A_1 & A_2 \\ B_1 & B_2 \end{vmatrix}\mathbf{k}$$

EXAMPLE 2.2 Given: $\mathbf{A} = 4\mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$ and $\mathbf{B} = 3\mathbf{i} + 5\mathbf{j} + 2\mathbf{k}$. Then

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} i & j & k \\ 4 & 2 & -3 \\ 3 & 5 & 2 \end{vmatrix} = 19\mathbf{i} - 17\mathbf{j} + 14\mathbf{k}$$

2.4 Triple Products

Dot and cross multiplication of three vectors **A**, **B**, and **C** may produce meaningful products, called *triple products*, of the form $(\mathbf{A} \cdot \mathbf{B})\mathbf{C}$, $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})$, and $\mathbf{A} \times (\mathbf{B} \times \mathbf{C})$.

The following proposition applies.

PROPOSITION 2.6: Suppose **A**, **B**, and **C** are vectors and m is a scalar. Then the following laws hold:

(i) In general, $(\mathbf{A} \cdot \mathbf{B})\mathbf{C} \neq \mathbf{A}(\mathbf{B} \cdot \mathbf{C})$.
(ii) $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B})$ = volume of a parallelepiped having **A**, **B**, and **C** as edges, or the negative of this volume, according as **A**, **B**, and **C** do or do not form a right-handed system.
(iii) In general, $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) \neq (\mathbf{A} \times \mathbf{B}) \times \mathbf{C}$ (Associative Law for Cross Products Fails)
(iv) $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \cdot \mathbf{C})\mathbf{B} - (\mathbf{A} \cdot \mathbf{B})\mathbf{C}$
$(\mathbf{A} \times \mathbf{B}) \times \mathbf{C} = (\mathbf{A} \cdot \mathbf{C})\mathbf{B} - (\mathbf{B} \cdot \mathbf{C})\mathbf{A}$

There is a simple formula for $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})$ when the unit vectors **i**, **j**, **k** are used.

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PROPOSITION 2.7: Given $\mathbf{A} = A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}$, $\mathbf{B} = B_1\mathbf{i} + B_2\mathbf{j} + B_3\mathbf{k}$, $\mathbf{C} = C_1\mathbf{i} + C_2\mathbf{j} + C_3\mathbf{k}$. Then

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \begin{vmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{vmatrix}$$

EXAMPLE 2.3 Given $\mathbf{A} = 4\mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$, $\mathbf{B} = 5\mathbf{i} + \mathbf{j} - 2\mathbf{k}$, $\mathbf{C} = 3\mathbf{i} - \mathbf{j} + 2\mathbf{k}$. Then:

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \begin{vmatrix} 4 & 2 & -3 \\ 5 & 1 & -2 \\ 3 & -1 & 2 \end{vmatrix} = 8 - 12 + 15 + 9 - 8 - 20 = -8.$$

2.5 Reciprocal Sets of Vectors

The sets $\mathbf{a}, \mathbf{b}, \mathbf{c}$ and $\mathbf{a}', \mathbf{b}', \mathbf{c}'$ are called *reciprocal sets* or *reciprocal systems* of vectors if:

$$\mathbf{a} \cdot \mathbf{a}' = \mathbf{b} \cdot \mathbf{b}' = \mathbf{c} \cdot \mathbf{c}' = 1$$

$$\mathbf{a}' \cdot \mathbf{b} = \mathbf{a}' \cdot \mathbf{c} = \mathbf{b}' \cdot \mathbf{a} = \mathbf{b}' \cdot \mathbf{c} = \mathbf{c}' \cdot \mathbf{a} = \mathbf{c}' \cdot \mathbf{b} = 0$$

That is, each vector is orthogonal to the reciprocal of the other two vectors in the system.

PROPOSITION 2.8: The sets $\mathbf{a}, \mathbf{b}, \mathbf{c}$ and $\mathbf{a}', \mathbf{b}', \mathbf{c}'$ are *reciprocal sets* of vectors if and only if

$$\mathbf{a}' = \frac{\mathbf{b} \times \mathbf{c}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}}, \quad \mathbf{b}' = \frac{\mathbf{c} \times \mathbf{a}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}}, \quad \mathbf{c}' = \frac{\mathbf{a} \times \mathbf{b}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}}$$

where $\mathbf{a} \cdot \mathbf{b} \times \mathbf{c} \neq 0$.

SOLVED PROBLEMS

Dot or Scalar Product

2.1. Prove Proposition 2.1(i): $\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A}$.

Solution

$$\mathbf{A} \cdot \mathbf{B} = |A||B| \cos\theta = |B||A| \cos\theta = \mathbf{B} \cdot \mathbf{A}.$$

Thus the commutative law for dot products is valid.

2.2. Prove that the projection of $\mathbf{A}$ on $\mathbf{B}$ is equal to $\mathbf{A} \cdot \mathbf{b}$ where $\mathbf{b}$ is a unit vector in the direction of $\mathbf{B}$.

Solution

Through the initial and terminal points of $\mathbf{A}$ pass planes perpendicular to $\mathbf{B}$ at G and H as in Fig. 2-1. Thus

$$\text{Projection of } \mathbf{A} \text{ on } \mathbf{B} = \overline{GH} = \overline{EF} = \mathbf{A}\cos\theta = \mathbf{A} \cdot \mathbf{b}$$

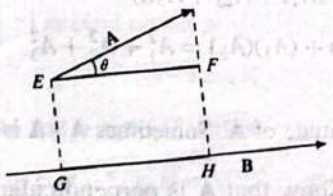


Fig. 2-1

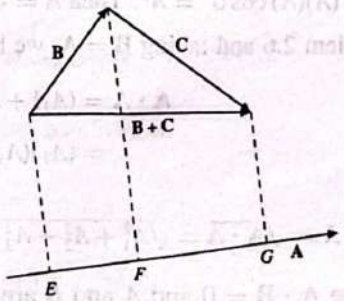


Fig. 2-2

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2.9. Find the angle between $\mathbf{A} = 2\mathbf{i} + 2\mathbf{j} - \mathbf{k}$ and $\mathbf{B} = 7\mathbf{i} + 24\mathbf{k}$.

Solution

We have $\mathbf{A} \cdot \mathbf{B} = |A||B| \cos\theta$.

$$|A| = \sqrt{(2)^2 + (2)^2 + (-1)^2} = 3 \quad \text{and} \quad |B| = \sqrt{(7)^2 + (0)^2 + (24)^2} = 25$$

$$\mathbf{A} \cdot \mathbf{B} = (2)(7) + (2)(0) + (-1)(24) = -10$$

Therefore,

$$\cos\theta = \frac{A \cdot B}{|A||B|} = \frac{-10}{(3)(25)} = \frac{-2}{15} = -0.1333 \text{ and } \theta = 98° \text{ (approximately).}$$

2.10. Determine the value of α so that $\mathbf{A} = 2\mathbf{i} + \alpha\mathbf{j} + \mathbf{k}$ and $\mathbf{B} = \mathbf{i} + 3\mathbf{j} - 8\mathbf{k}$ are perpendicular.

Solution

By Proposition 2.1(v), **A** and **B** are perpendicular when $\mathbf{A} \cdot \mathbf{B} = 0$. Thus,

$$\mathbf{A} \cdot \mathbf{B} = (2)(1) + (\alpha)(3) + (1)(-8) = 2 + 3\alpha - 8 = 0$$

and if $\alpha = 2$.

2.11. Show that the vectors $\mathbf{A} = -\mathbf{i} + \mathbf{j}$, $\mathbf{B} = -\mathbf{i} - \mathbf{j} - 2\mathbf{k}$, $\mathbf{C} = 2\mathbf{j} + 2\mathbf{k}$ form a right triangle.

Solution

First we show that the vectors form a triangle. From Fig. 2-3, we see that the vectors form a triangle if:

(a) one of the vectors, say (3), is the sum of (1) and (2) or
(b) the sum of the vectors (1) + (2) + (3) is zero

according as (a) two vectors have a common terminal point, or (b) none of the vectors have a common terminal point. By trial, we find $\mathbf{A} = \mathbf{B} + \mathbf{C}$ so the vectors do form a triangle.

Since $\mathbf{A} \cdot \mathbf{B} = (-1)(-1) + (1)(-1) + (0)(-2) = 0$, it follows that **A** and **B** are perpendicular and the triangle is a right triangle.

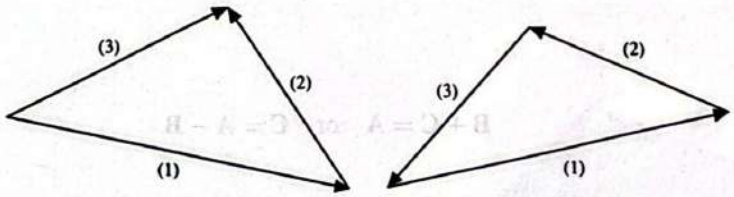


Fig. 2-3

2.12. Find the angles that the vector $\mathbf{A} = 4\mathbf{i} - 8\mathbf{j} + \mathbf{k}$ makes with the coordinate axes.

Solution

Let α, β, γ be the angles that **A** makes with the positive x, y, z axes, respectively.

$$\mathbf{A} \cdot \mathbf{i} = |A|(1)\cos\alpha = \sqrt{(4)^2 + (-8)^2 + (1)^2}\cos\alpha = 9\cos\alpha$$

$$\mathbf{A} \cdot \mathbf{i} = (4\mathbf{i} - 8\mathbf{j} + \mathbf{k}) \cdot \mathbf{i} = 4$$

Then $\cos\alpha = 4/9 = 0.4444$ and $\alpha = 63.6°$ approximately. Similarly,

$$\cos\beta = -8/9,\ \beta = 152.7° \quad \text{and} \quad \cos\gamma = 1/9,\ \gamma = 83.6°$$

The cosines of α, β, γ are called the *direction cosines* of the vector **A**.

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2.13. Find the projection of the vector $\mathbf{A} = \mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$ on the vector $\mathbf{B} = \mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$.

Solution

We use the result of Problem 2.2. A unit vector in the direction of $\mathbf{B}$ is

$$\mathbf{b} = \mathbf{B}/|B| = (\mathbf{i} + 2\mathbf{j} + 2\mathbf{k})/\sqrt{1+4+4} = \mathbf{i}/3 + 2\mathbf{j}/3 + 2\mathbf{k}/3$$

The projection of $\mathbf{A}$ on vector $\mathbf{B}$ is

$$\mathbf{A} \cdot \mathbf{b} = (\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}) \cdot (\mathbf{i}/3 + 2\mathbf{j}/3 + 2\mathbf{k}/3) = (1)(1/3) + (-2)(2/3) + (3)(2/3) = 1.$$

2.14. Without making use of the cross product, determine a unit vector perpendicular to the plane of $\mathbf{A} = 2\mathbf{i} - 6\mathbf{j} - 3\mathbf{k}$ and $\mathbf{B} = 4\mathbf{i} + 3\mathbf{j} - \mathbf{k}$.

Solution

Let vector $\mathbf{C} = c_1\mathbf{i} + c_2\mathbf{j} + c_3\mathbf{k}$ be perpendicular to the plane of $\mathbf{A}$ and $\mathbf{B}$. Then $\mathbf{C}$ is perpendicular to $\mathbf{A}$ and also to $\mathbf{B}$. Hence,

$$\mathbf{C} \cdot \mathbf{A} = 2c_1 - 6c_2 - 3c_3 = 0 \quad \text{or} \quad (1)\ 2c_1 - 6c_2 = 3c_3$$
$$\mathbf{C} \cdot \mathbf{B} = 4c_1 + 3c_2 - c_3 = 0 \quad \text{or} \quad (2)\ 4c_1 + 3c_2 = c_3$$

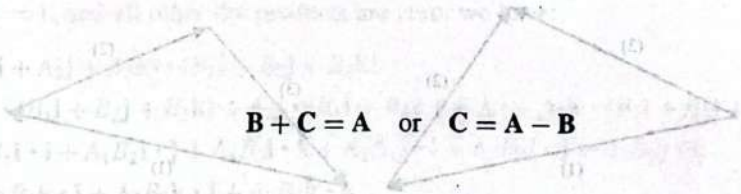
Solving (1) and (2) simultaneously: $c_1 = \frac{1}{2}c_3$, $c_2 = -\frac{1}{3}c_3$, $\mathbf{C} = c_3\left(\frac{1}{2}\mathbf{i} - \frac{1}{3}\mathbf{j} + \mathbf{k}\right)$.

Then a unit vector in the direction of $\mathbf{C}$ is $\dfrac{\mathbf{C}}{|C|} = \dfrac{c_3\left(\frac{1}{2}\mathbf{i} - \frac{1}{3}\mathbf{j} + \mathbf{k}\right)}{\sqrt{c_3^2\left[\left(\frac{1}{2}\right)^2 + \left(-\frac{1}{3}\right)^2 + (1)^2\right]}} = \pm\left(\frac{3}{7}\mathbf{i} - \frac{2}{7}\mathbf{j} + \frac{6}{7}\mathbf{k}\right)$.

2.15. Prove the law of cosines for plane triangles.

Solution

From Fig. 2-4,



$$\mathbf{B} + \mathbf{C} = \mathbf{A} \quad \text{or} \quad \mathbf{C} = \mathbf{A} - \mathbf{B}$$

Then

$$\mathbf{C} \cdot \mathbf{C} = (\mathbf{A} - \mathbf{B}) \cdot (\mathbf{A} - \mathbf{B}) = \mathbf{A} \cdot \mathbf{A} + \mathbf{B} \cdot \mathbf{B} - 2\mathbf{A} \cdot \mathbf{B}$$

and

$$C^2 = A^2 + B^2 - 2AB\cos\theta$$

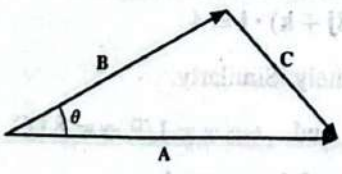


Fig. 2-4

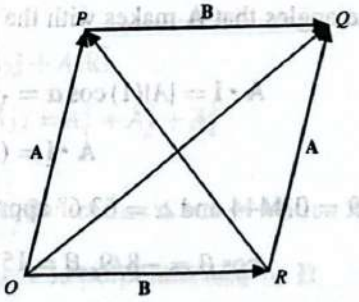


Fig. 2-5

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Also, since $\mathbf{B}+\mathbf{C}=\mathbf{B}_\perp+\mathbf{B}_\parallel+\mathbf{C}_\perp+\mathbf{C}_\parallel=(\mathbf{B}_\perp+\mathbf{C}_\perp)+(\mathbf{B}_\parallel+\mathbf{C}_\parallel)$ it follows that

$$\mathbf{A}\times(\mathbf{B}_\perp+\mathbf{C}_\perp)=\mathbf{A}\times(\mathbf{B}+\mathbf{C}).$$

Now $\mathbf{B}_\perp$ and $\mathbf{C}_\perp$ are vectors perpendicular to $\mathbf{A}$ and so by Problem 2.25,

$$\mathbf{A}\times(\mathbf{B}_\perp+\mathbf{C}_\perp)=\mathbf{A}\times\mathbf{B}_\perp+\mathbf{A}\times\mathbf{C}_\perp$$

Then

$$\mathbf{A}\times(\mathbf{B}+\mathbf{C})=\mathbf{A}\times\mathbf{B}+\mathbf{A}\times\mathbf{C}$$

and the distributive law holds. Multiplying by -1, using Problem 2.21, this becomes $(\mathbf{B}+\mathbf{C})\times\mathbf{A}=\mathbf{B}\times\mathbf{A}+\mathbf{C}\times\mathbf{A}$. Note that the order of factors in cross products is important. The usual laws of algebra apply only if proper order is maintained.

2.27. Suppose $\mathbf{A}=A_1\mathbf{i}+A_2\mathbf{j}+A_3\mathbf{k}$ and $\mathbf{B}=B_1\mathbf{i}+B_2\mathbf{j}+B_3\mathbf{k}$. Prove $\mathbf{A}\times\mathbf{B}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\A_1&A_2&A_3\\B_1&B_2&B_3\end{vmatrix}$.

Solution

$$\begin{aligned}\mathbf{A}\times\mathbf{B}&=(A_1\mathbf{i}+A_2\mathbf{j}+A_3\mathbf{k})\times(B_1\mathbf{i}+B_2\mathbf{j}+B_3\mathbf{k})\\&=A_1\mathbf{i}\times(B_1\mathbf{i}+B_2\mathbf{j}+B_3\mathbf{k})+A_2\mathbf{j}\times(B_1\mathbf{i}+B_2\mathbf{j}+B_3\mathbf{k})+A_3\mathbf{k}\times(B_1\mathbf{i}+B_2\mathbf{j}+B_3\mathbf{k})\\&=A_1B_1\mathbf{i}\times\mathbf{i}+A_1B_2\mathbf{i}\times\mathbf{j}+A_1B_3\mathbf{i}\times\mathbf{k}+A_2B_1\mathbf{j}\times\mathbf{i}+A_2B_2\mathbf{j}\times\mathbf{j}+A_2B_3\mathbf{j}\times\mathbf{k}\\&\quad+A_3B_1\mathbf{k}\times\mathbf{i}+A_3B_2\mathbf{k}\times\mathbf{j}+A_3B_3\mathbf{k}\times\mathbf{k}\\&=(A_2B_3-A_3B_2)\mathbf{i}+(A_3B_1-A_1B_3)\mathbf{j}+(A_1B_2-A_2B_1)\mathbf{k}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\A_1&A_2&A_3\\B_1&B_2&B_3\end{vmatrix}.\end{aligned}$$

2.28. Suppose $\mathbf{A}=\mathbf{j}+2\mathbf{k}$ and $\mathbf{B}=\mathbf{i}+2\mathbf{j}+3\mathbf{k}$. Find: (a) $\mathbf{A}\times\mathbf{B}$, (b) $\mathbf{B}\times\mathbf{A}$, (c) $(\mathbf{A}+\mathbf{B})\times(\mathbf{A}-\mathbf{B})$.

Solution

(a) $$\mathbf{A}\times\mathbf{B}=(\mathbf{j}+2\mathbf{k})\times(\mathbf{i}+2\mathbf{j}+3\mathbf{k})=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\0&1&2\\1&2&3\end{vmatrix}$$

$$=\begin{vmatrix}1&2\\2&3\end{vmatrix}\mathbf{i}-\begin{vmatrix}0&2\\1&3\end{vmatrix}\mathbf{j}+\begin{vmatrix}0&1\\1&2\end{vmatrix}\mathbf{k}=-\mathbf{i}+2\mathbf{j}-\mathbf{k}.$$

(b) $$\mathbf{B}\times\mathbf{A}=(\mathbf{i}+2\mathbf{j}+3\mathbf{k})\times(\mathbf{j}+2\mathbf{k})=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\1&2&3\\0&1&2\end{vmatrix}$$

$$=\begin{vmatrix}2&3\\1&2\end{vmatrix}\mathbf{i}-\begin{vmatrix}1&3\\0&2\end{vmatrix}\mathbf{j}+\begin{vmatrix}1&2\\0&1\end{vmatrix}\mathbf{k}=\mathbf{i}-2\mathbf{j}+\mathbf{k}.$$

Comparing with (a), we have $\mathbf{A}\times\mathbf{B}=-(\mathbf{B}\times\mathbf{A})$. Note this is equivalent to the theorem: If two rows of a determinant are interchanged, the determinant changes sign.

(c) $\mathbf{A}+\mathbf{B}=\mathbf{i}+3\mathbf{j}+5\mathbf{k}$ and $\mathbf{A}-\mathbf{B}=-\mathbf{i}-\mathbf{j}-\mathbf{k}$. Then

$$(\mathbf{A}+\mathbf{B})\times(\mathbf{A}-\mathbf{B})=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\1&3&5\\-1&-1&-1\end{vmatrix}=\begin{vmatrix}3&5\\-1&-1\end{vmatrix}\mathbf{i}-\begin{vmatrix}1&5\\-1&-1\end{vmatrix}\mathbf{j}+\begin{vmatrix}1&3\\-1&-1\end{vmatrix}\mathbf{k}$$

$$=2\mathbf{i}-4\mathbf{j}+2\mathbf{k}.$$

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The vectors associated with each of the faces of the tetrahedron are

$$\mathbf{V}_1 = \tfrac{1}{2}\mathbf{A}\times\mathbf{B},\quad \mathbf{V}_2 = \tfrac{1}{2}\mathbf{B}\times\mathbf{C},\quad \mathbf{V}_3 = \tfrac{1}{2}\mathbf{C}\times\mathbf{A},\quad \mathbf{V}_4 = \tfrac{1}{2}(\mathbf{C}-\mathbf{A})\times(\mathbf{B}-\mathbf{A})$$

Then

$$\begin{aligned}\mathbf{V}_1+\mathbf{V}_2+\mathbf{V}_3+\mathbf{V}_4 &= \tfrac{1}{2}[\mathbf{A}\times\mathbf{B}+\mathbf{B}\times\mathbf{C}+\mathbf{C}\times\mathbf{A}+(\mathbf{C}-\mathbf{A})\times(\mathbf{B}-\mathbf{A})]\\ &= \tfrac{1}{2}[\mathbf{A}\times\mathbf{B}+\mathbf{B}\times\mathbf{C}+\mathbf{C}\times\mathbf{A}+\mathbf{C}\times\mathbf{B}-\mathbf{C}\times\mathbf{A}-\mathbf{A}\times\mathbf{B}+\mathbf{A}\times\mathbf{A}] = 0.\end{aligned}$$

This result can be generalized to closed polyhedra and in the limiting case to any closed surface.

Because of the application presented here, it is sometimes convenient to assign a direction to area and we speak of the *vector area*.

2.33. Find the area of the triangle having vertices at $P(1, 3, 2)$, $Q(2, -1, 1)$, $R(-1, 2, 3)$.

Solution

$$\mathbf{PQ} = (2-1)\mathbf{i}+(-1-3)\mathbf{j}+(1-2)\mathbf{k} = \mathbf{i}-4\mathbf{j}-\mathbf{k}$$
$$\mathbf{PR} = (-1-1)\mathbf{i}+(2-3)\mathbf{j}+(3-2)\mathbf{k} = -2\mathbf{i}-\mathbf{j}+\mathbf{k}$$

From Problem 2.30,

$$\text{area of triangle} = \frac{1}{2}|\mathbf{PQ}\times\mathbf{PR}| = \frac{1}{2}|(\mathbf{i}-4\mathbf{j}-\mathbf{k})\times(-2\mathbf{i}-\mathbf{j}+\mathbf{k})|$$
$$= \frac{1}{2}\begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\ 1 & -4 & -1\\ -2 & -1 & 1\end{vmatrix} = \frac{1}{2}|-5\mathbf{i}+\mathbf{j}-9\mathbf{k}| = \frac{1}{2}\sqrt{(-5)^2+(1)^2+(-9)^2} = \frac{1}{2}\sqrt{107}.$$

2.34. Determine a unit vector perpendicular to the plane of $\mathbf{A} = 2\mathbf{i}-6\mathbf{j}-3\mathbf{k}$ and $\mathbf{B} = 4\mathbf{i}+3\mathbf{j}-\mathbf{k}$.

Solution

$\mathbf{A}\times\mathbf{B}$ is a vector perpendicular to the plane of $\mathbf{A}$ and $\mathbf{B}$.

$$\mathbf{A}\times\mathbf{B} = \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\ 2 & -6 & -3\\ 4 & 3 & -1\end{vmatrix} = 15\mathbf{i}-10\mathbf{j}+30\mathbf{k}$$

A unit vector parallel to $\mathbf{A}\times\mathbf{B}$ is $\dfrac{\mathbf{A}\times\mathbf{B}}{|\mathbf{A}\times\mathbf{B}|} = \dfrac{15\mathbf{i}-10\mathbf{j}+30\mathbf{k}}{\sqrt{(15)^2+(-10)^2+(30)^2}} = \dfrac{3}{7}\mathbf{i}-\dfrac{2}{7}\mathbf{j}+\dfrac{6}{7}\mathbf{k}$.

Another unit vector, opposite in direction, is $(-3\mathbf{i}+2\mathbf{j}-6\mathbf{k})/7$. Compare with Problem 2.14.

2.35. Find an expression for the moment of a force $\mathbf{F}$ about a point P as in Fig. 2-14.

Solution

The moment $\mathbf{M}$ of $\mathbf{F}$ about P is in magnitude equal to P to the line of action of $\mathbf{F}$. Then, if $\mathbf{r}$ is the vector from P to the initial point Q of $\mathbf{F}$,

$$M = F(r\sin\theta) = rF\sin\theta = |\mathbf{r}\times\mathbf{F}|$$

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2.54. Show that any vector $\mathbf{r}$ can be expressed in terms of the reciprocal vectors of Problem 2.53 as

$$\mathbf{r} = (\mathbf{r}\cdot\mathbf{a}')\mathbf{a} + (\mathbf{r}\cdot\mathbf{b}')\mathbf{b} + (\mathbf{r}\cdot\mathbf{c}')\mathbf{c}.$$

Solution

From Problem 2.50, $\mathbf{B}(\mathbf{A}\cdot\mathbf{C}\times\mathbf{D}) - \mathbf{A}(\mathbf{B}\cdot\mathbf{C}\times\mathbf{D}) = \mathbf{C}(\mathbf{A}\cdot\mathbf{B}\times\mathbf{D}) - \mathbf{D}(\mathbf{A}\cdot\mathbf{B}\times\mathbf{C})$. Then

$$\mathbf{D} = \frac{\mathbf{A}(\mathbf{B}\cdot\mathbf{C}\times\mathbf{D})}{\mathbf{A}\cdot\mathbf{B}\times\mathbf{C}} - \frac{\mathbf{B}(\mathbf{A}\cdot\mathbf{C}\times\mathbf{D})}{\mathbf{A}\cdot\mathbf{B}\times\mathbf{C}} + \frac{\mathbf{C}(\mathbf{A}\cdot\mathbf{B}\times\mathbf{D})}{\mathbf{A}\cdot\mathbf{B}\times\mathbf{C}}$$

Let $\mathbf{A} = \mathbf{a}$, $\mathbf{B} = \mathbf{b}$, $\mathbf{C} = \mathbf{c}$, and $\mathbf{D} = \mathbf{r}$. Then

$$\begin{aligned}\mathbf{r} &= \frac{\mathbf{r}\cdot\mathbf{b}\times\mathbf{c}}{\mathbf{a}\cdot\mathbf{b}\times\mathbf{c}}\mathbf{a} + \frac{\mathbf{r}\cdot\mathbf{c}\times\mathbf{a}}{\mathbf{a}\cdot\mathbf{b}\times\mathbf{c}}\mathbf{b} + \frac{\mathbf{r}\cdot\mathbf{a}\times\mathbf{b}}{\mathbf{a}\cdot\mathbf{b}\times\mathbf{c}}\mathbf{c} \\ &= \mathbf{r}\cdot\left(\frac{\mathbf{b}\times\mathbf{c}}{\mathbf{a}\cdot\mathbf{b}\times\mathbf{c}}\right)\mathbf{a} + \mathbf{r}\cdot\left(\frac{\mathbf{c}\times\mathbf{a}}{\mathbf{a}\cdot\mathbf{b}\times\mathbf{c}}\right)\mathbf{b} + \mathbf{r}\cdot\left(\frac{\mathbf{a}\times\mathbf{b}}{\mathbf{a}\cdot\mathbf{b}\times\mathbf{c}}\right)\mathbf{c} \\ &= (\mathbf{r}\cdot\mathbf{a}')\mathbf{a} + (\mathbf{r}\cdot\mathbf{b}')\mathbf{b} + (\mathbf{r}\cdot\mathbf{c}')\mathbf{c}.\end{aligned}$$

SUPPLEMENTARY PROBLEMS

2.55. Evaluate: (a) $\mathbf{k}\cdot(\mathbf{i}+\mathbf{j})$, (b) $(\mathbf{i}-2\mathbf{k})\cdot(\mathbf{j}+3\mathbf{k})$, (c) $(2\mathbf{i}-\mathbf{j}+3\mathbf{k})\cdot(3\mathbf{i}+2\mathbf{j}-\mathbf{k})$.

2.56. Suppose $\mathbf{A} = \mathbf{i}+3\mathbf{j}-2\mathbf{k}$ and $\mathbf{B} = 4\mathbf{i}-2\mathbf{j}+4\mathbf{k}$. Find: (a) $\mathbf{A}\cdot\mathbf{B}$, (b) A, (c) B, (d) $|3\mathbf{A}+2\mathbf{B}|$, (e) $(2\mathbf{A}+\mathbf{B})\cdot(\mathbf{A}-2\mathbf{B})$.

2.57. Find the angle between (a) $\mathbf{A} = 3\mathbf{i}+2\mathbf{j}-6\mathbf{k}$ and $\mathbf{B} = 4\mathbf{i}-3\mathbf{j}+\mathbf{k}$; (b) $\mathbf{C} = 4\mathbf{i}-2\mathbf{j}+4\mathbf{k}$ and $\mathbf{D} = 3\mathbf{i}-6\mathbf{j}-2\mathbf{k}$.

2.58. Find the values of a for which vectors $\mathbf{A}$ and $\mathbf{B}$ are perpendicular where:
(a) $\mathbf{A} = a\mathbf{i}-2\mathbf{j}+\mathbf{k}$ and $\mathbf{B} = 2a\mathbf{i}+a\mathbf{j}-4\mathbf{k}$, (b) $\mathbf{A} = 2\mathbf{i}+\mathbf{j}+a\mathbf{k}$ and $\mathbf{B} = 2\mathbf{i}+a\mathbf{j}+\mathbf{k}$.

2.59. Find the acute angles that the line joining the points (1, −3, 2) and (3, −5, 1) makes with the coordinate axes.

2.60. Find the direction cosines of the line joining the points:
(a) (3, 2, −4) and (1, −1, 2), (b) (−5, 3, 3) and (−2, 7, 15).

2.61. Determine the angles of a triangle where two sides of a triangle are formed by the vectors:
(a) $\mathbf{A} = 3\mathbf{i}-4\mathbf{j}-\mathbf{k}$ and $\mathbf{B} = 4\mathbf{i}-\mathbf{j}+3\mathbf{k}$, (b) $\mathbf{A} = -2\mathbf{i}+5\mathbf{j}+6\mathbf{k}$ and $\mathbf{B} = 3\mathbf{i}+\mathbf{j}+2\mathbf{k}$.

2.62. The diagonals of a parallelogram are given by $\mathbf{A} = 3\mathbf{i}-4\mathbf{j}-\mathbf{k}$ and $\mathbf{B} = 2\mathbf{i}+3\mathbf{j}-6\mathbf{k}$. Show that the parallelogram is a rhombus and determine the length of its sides and angles.

2.63. Find the projection of the vector $\mathbf{A}$ on the vector $\mathbf{B}$ where:
(a) $\mathbf{A} = 2\mathbf{i}-3\mathbf{j}+6\mathbf{k}$ and $\mathbf{B} = \mathbf{i}+2\mathbf{j}+2\mathbf{k}$, (b) $\mathbf{A} = 2\mathbf{i}+\mathbf{j}-\mathbf{k}$ and $\mathbf{B} = -6\mathbf{i}+2\mathbf{j}-3\mathbf{k}$.

2.64. Find the projection of the vector $\mathbf{A} = 4\mathbf{i}-3\mathbf{j}+\mathbf{k}$ on the line passing through the points (2, 3, −1) and (−2, −4, 3).

2.65. Find a unit vector perpendicular to both vector $\mathbf{A}$ and vector $\mathbf{B}$ where:
(a) $\mathbf{A} = 4\mathbf{i}-\mathbf{j}+3\mathbf{k}$ and $\mathbf{B} = -2\mathbf{i}+\mathbf{j}-2\mathbf{k}$, (b) $\mathbf{A} = 6\mathbf{i}+22\mathbf{j}-5\mathbf{k}$ and $\mathbf{B} = \mathbf{i}+6\mathbf{j}-2\mathbf{k}$.

2.66. Find the acute angle formed by two diagonals of a cube.

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2.67. Find a unit vector parallel to the xy-plane and perpendicular to the vector $4\mathbf{i} - 3\mathbf{j} + \mathbf{k}$.

2.68. Show that $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ are mutually orthogonal unit vectors where:

(a) $\mathbf{A} = (2\mathbf{i} - 2\mathbf{j} + \mathbf{k})/3$, $\mathbf{B} = (\mathbf{i} + 2\mathbf{j} + 2\mathbf{k})/3$, and $\mathbf{C} = (2\mathbf{i} + \mathbf{j} - 2\mathbf{k})/3$

(b) $\mathbf{A} = (12\mathbf{i} - 4\mathbf{j} - 3\mathbf{k})/13$, $\mathbf{B} = (4\mathbf{i} + 3\mathbf{j} + 12\mathbf{k})/13$, and $\mathbf{C} = (3\mathbf{i} + 12\mathbf{j} - 4\mathbf{k})/13$.

2.69. Find the work done in moving an object along a straight line:

(a) from $(3, 2, -1)$ to $(2, -1, 4)$ in a force field given by $\mathbf{F} = 4\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$.

(b) from $(3, 4, 5)$ to $(-1, 9, 9)$ in a force field given by $\mathbf{F} = -3\mathbf{i} + 5\mathbf{j} - 6\mathbf{k}$.

2.70. Let $\mathbf{F}$ be a constant vector field force. Show that the work done in moving an object around any closed polygon in this force field is zero.

2.71. Prove that an angle inscribed in a semicircle is a right angle.

2.72. Let $ABCD$ be a parallelogram. Prove that $\overline{AB}^2 + \overline{BC}^2 + \overline{CD}^2 + \overline{DA}^2 = \overline{AC}^2 + \overline{BD}^2$

2.73. Let $ABCD$ be any quadrilateral where P and Q are the midpoints of its diagonal. Prove that

$$\overline{AB}^2 + \overline{BC}^2 + \overline{CD}^2 + \overline{DA}^2 = \overline{AC}^2 + \overline{BD}^2 + 4\overline{PQ}^2$$

This is a generalization of the preceding problem.

2.74. Consider a plane P perpendicular to a given vector $\mathbf{A}$ and distance p from the origin. (a) Find an equation of the plane P. (b) Express the equation in (a) in rectangular coordinates.

2.75. Let $\mathbf{r}_1$ and $\mathbf{r}_2$ be unit vectors in the xy-plane making angles α and β with the positive x-axis.

(a) Prove that $\mathbf{r}_1 = \cos\alpha\mathbf{i} + \sin\alpha\mathbf{j}$ and $\mathbf{r}_2 = \cos\beta\mathbf{i} + \sin\beta\mathbf{j}$.

(b) By considering $\mathbf{r}_1 \cdot \mathbf{r}_2$, prove the trigonometric formulas

$$\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta \quad \text{and} \quad \cos(\alpha + \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$$

2.76. Let $\mathbf{a}$ be the position vector of a given point (x_1, y_1, z_1), and let $\mathbf{r}$ be the position vector of any point (x, y, z). Describe the locus of $\mathbf{r}$ if: (a) $|\mathbf{r} - \mathbf{a}| = 3$, (b) $(\mathbf{r} - \mathbf{a}) \cdot \mathbf{a} = 0$, (c) $(\mathbf{r} - \mathbf{a}) \cdot \mathbf{r} = 0$.

2.77. Suppose $\mathbf{A} = 3\mathbf{i} + \mathbf{j} + 2\mathbf{k}$ and $\mathbf{B} = \mathbf{i} - 2\mathbf{j} - 4\mathbf{k}$ are the position vectors of points P and Q, respectively.

(a) Find an equation for the plane passing through Q and perpendicular to the line PQ.

(b) Find the distance from the point $(-1, 1, 1)$ to the plane.

2.78. Evaluate each of the following: (a) $2\mathbf{j} \times (3\mathbf{i} - 4\mathbf{k})$, (b) $(\mathbf{i} + 2\mathbf{j}) \times \mathbf{k}$, (c) $(2\mathbf{i} - 4\mathbf{k}) \times (\mathbf{i} + 2\mathbf{j})$, (d) $(4\mathbf{i} + \mathbf{j} - 2\mathbf{k}) \times (3\mathbf{i} + \mathbf{k})$, (e) $(2\mathbf{i} + \mathbf{j} - \mathbf{k}) \times (3\mathbf{i} - 2\mathbf{j} + 4\mathbf{k})$.

2.79. Suppose $\mathbf{A} = 3\mathbf{i} - \mathbf{j} - 2\mathbf{k}$ and $\mathbf{B} = 2\mathbf{i} + 3\mathbf{j} + \mathbf{k}$. Find: (a) $|\mathbf{A} \times \mathbf{B}|$, (b) $(\mathbf{A} + 2\mathbf{B}) \times (2\mathbf{A} - \mathbf{B})$, (c) $|(\mathbf{A} + \mathbf{B}) \times (\mathbf{A} - \mathbf{B})|$.

2.80. Suppose $\mathbf{A} = \mathbf{i} - 2\mathbf{j} - 3\mathbf{k}$, $\mathbf{B} = 2\mathbf{i} + \mathbf{j} - \mathbf{k}$, $\mathbf{C} = \mathbf{i} + 3\mathbf{j} - 2\mathbf{k}$. Find:

(a) $|(\mathbf{A} \times \mathbf{B}) \times \mathbf{C}|$ (c) $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})$, (e) $(\mathbf{A} \times \mathbf{B}) \times (\mathbf{B} \times \mathbf{C})$

(b) $|\mathbf{A} \times (\mathbf{B} \times \mathbf{C})|$ (d) $(\mathbf{A} \times \mathbf{B}) \cdot \mathbf{C}$, (f) $(\mathbf{A} \times \mathbf{B})(\mathbf{B} \cdot \mathbf{C})$

2.81. Suppose $\mathbf{A} \neq \mathbf{0}$ and both of the following conditions hold simultaneously: (a) $\mathbf{A} \cdot \mathbf{B} = \mathbf{A} \cdot \mathbf{C}$, and (b) $\mathbf{A} \times \mathbf{B} = \mathbf{A} \times \mathbf{C}$. Show that $\mathbf{B} = \mathbf{C}$ but, if only one of the conditions holds, then $\mathbf{B} \neq \mathbf{C}$ necessarily.

2.82. Find the area of a parallelogram having diagonals: (a) $\mathbf{A} = 3\mathbf{i} + \mathbf{j} - 2\mathbf{k}$ and $\mathbf{B} = \mathbf{i} - 3\mathbf{j} - 4\mathbf{k}$, (b) $\mathbf{A} = 2\mathbf{i} + 4\mathbf{j}$ and $\mathbf{B} = -4\mathbf{i} + 4\mathbf{k}$.

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CHAPTER 3

Vector Differentiation

3.1 Introduction

The reader is familiar with the differentiation of real valued functions $f(x)$ of one variable. Specifically, we have:

$$f'(x) = \frac{df}{dx} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Here we extend this definition to vector-valued functions of a single variable.

3.2 Ordinary Derivatives of Vector-Valued Functions

Suppose $\mathbf{R}(u)$ is a vector depending on a single scalar variable u. Then

$$\frac{\Delta \mathbf{R}}{\Delta u} = \frac{\mathbf{R}(u + \Delta u) - \mathbf{R}(u)}{\Delta u}$$

where Δu denotes an increment in u as shown in Fig. 3-1.

The ordinary derivative of the vector $\mathbf{R}(u)$ with respect to the scalar u is given as follows when the limit exists:

$$\frac{d\mathbf{R}}{du} = \lim_{\Delta u \to 0} \frac{\Delta \mathbf{R}}{\Delta u} = \lim_{\Delta u \to 0} \frac{\mathbf{R}(u + \Delta u) - \mathbf{R}(u)}{\Delta u}$$

Since $d\mathbf{R}/du$ is itself a vector depending on u, we can consider its derivative with respect to u. If this derivative exists, we denote it by $d^2\mathbf{R}/du^2$. Similarly, higher-order derivatives are described.

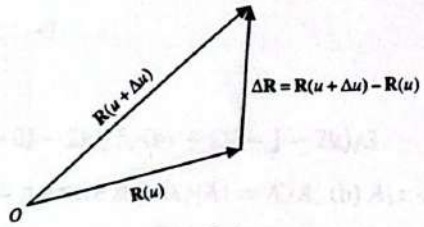


Fig. 3-1

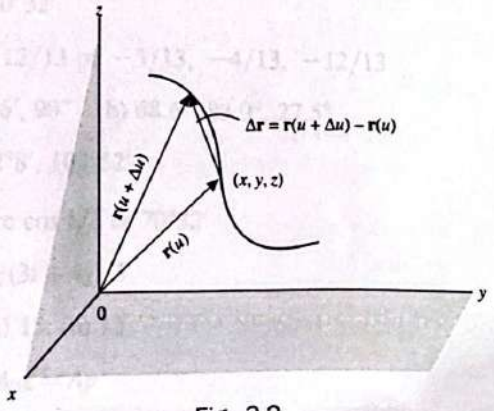


Fig. 3-2

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Space Curves

Consider now the position vector $\mathbf{r}(u)$ joining the origin O of a coordinate system and any point (x, y, z). Then

$$\mathbf{r}(u) = x(u)\mathbf{i} + y(u)\mathbf{j} + z(u)\mathbf{k}$$

and the specification of the vector function $\mathbf{r}(u)$ defines x, y, and z as functions of u.

As u changes, the terminal point of $\mathbf{r}$ describes a *space curve* having parametric equations

$$x = x(u), \quad y = y(u), \quad z = z(u)$$

Then the following is a vector in the direction of $\Delta\mathbf{r}$ if $\Delta u > 0$ and in the direction of $-\Delta\mathbf{r}$ if $\Delta u < 0$ [as pictured in Fig. 3-2]:

$$\frac{\Delta\mathbf{r}}{\Delta u} = \frac{\mathbf{r}(u + \Delta u) - \mathbf{r}(u)}{\Delta u}$$

Suppose

$$\lim_{\Delta u \to 0} \frac{\Delta\mathbf{r}}{\Delta u} = \frac{d\mathbf{r}}{du}$$

exists. Then the limit will be a vector in the direction of the tangent to the space curve at (x, y, z) and it is given by

$$\frac{d\mathbf{r}}{du} = \frac{dx}{du}\mathbf{i} + \frac{dy}{du}\mathbf{j} + \frac{dz}{du}\mathbf{k}$$

Motion: Velocity and Acceleration

Suppose a particle P moves along a space curve C whose parametric equations are $x = x(t)$, $y = y(t)$, $z = z(t)$, where t represents time. Then the position vector of the particle P along the curve is

$$\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$$

In such a case, the velocity $\mathbf{v}$ and acceleration $\mathbf{a}$ of the particle P is given by:

$$\mathbf{v} = \mathbf{v}(t) = \frac{d\mathbf{r}}{dt} = \frac{dx}{dt}\mathbf{i} + \frac{dy}{dt}\mathbf{j} + \frac{dz}{dt}\mathbf{k}$$

$$\mathbf{a} = \mathbf{a}(t) = \frac{d^2\mathbf{r}}{dt^2} = \frac{d\mathbf{v}}{dt} = \frac{d^2x}{dt^2}\mathbf{i} + \frac{d^2y}{dt^2}\mathbf{j} + \frac{d^2z}{dt^2}\mathbf{k}$$

EXAMPLE 3.1 Suppose a particle P moves along a curve whose parametric equations, where t is time, follows:

$$x = 40t^2 + 8t, \quad y = 2\cos 3t, \quad z = 2\sin 3t$$

(a) Determine its velocity and acceleration at any time.
(b) Find the magnitudes of the velocity and acceleration at $t = 0$.

(a) The position vector of the particle P is

$$\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k} = (40t^2 + 8t)\mathbf{i} + (2\cos 3t)\mathbf{j} + (2\sin 3t)\mathbf{k}$$

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Then the velocity $\mathbf{v}$ and acceleration $\mathbf{a}$ of P follow:

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = (80t + 8)\mathbf{i} + (-6\sin 3t)\mathbf{j} + (6\cos 3t)\mathbf{k}$$

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = 80\mathbf{i} + (-18\cos 3t)\mathbf{j} + (-18\sin 3t)\mathbf{k}.$$

(b) At $t = 0$, $\mathbf{v} = 8\mathbf{i} + 6\mathbf{k}$, and $\mathbf{a} = 80\mathbf{i} - 18\mathbf{j}$. Magnitudes of velocity $\mathbf{v}$ and acceleration $\mathbf{a}$ follow:

$$|\mathbf{v}| = \sqrt{(8)^2 + (6)^2} = 10 \quad \text{and} \quad |\mathbf{a}| = \sqrt{(80)^2 + (-18)^2} = 82$$

3.3 Continuity and Differentiability

A scalar function $\phi(u)$ is called *continuous* at u if

$$\lim_{\Delta u \to 0} \phi(u + \Delta u) = \phi(u)$$

Equivalently, $\phi(u)$ is continuous at u if, for each positive number ϵ, we can find a positive number δ such that

$$|\phi(u + \Delta u) - \phi(u)| < \epsilon \quad \text{whenever } |\Delta u| < \delta$$

A vector function $\mathbf{R}(u) = R_1(u)\mathbf{i} + R_2(u)\mathbf{j} + R_3(u)\mathbf{k}$ is called *continuous* at u if the three functions $R_1(u)$, $R_2(u)$, $R_3(u)$ are continuous at u or if $\lim_{\Delta u \to 0} \mathbf{R}(u + \Delta u) = \mathbf{R}(u)$. Equivalently, $\mathbf{R}(u)$ is continuous at u if, for each positive number ϵ, we can find a positive number δ such that

$$|\mathbf{R}(u + \Delta u) - \mathbf{R}(u)| < \epsilon \quad \text{whenever } |\Delta u| < \delta$$

A scalar or vector function of u is called *differentiable of order n* if its nth derivative exists. A function that is differentiable is necessarily continuous but the converse is not true. Unless otherwise stated, we assume that all functions considered are differentiable to any order needed in a particular discussion.

The following proposition applies.

PROPOSITION 2.1 Suppose $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ are differentiable vector functions of a scalar u, and ϕ is a differentiable scalar function of u. Then the following laws hold:

(i) $\dfrac{d}{du}(\mathbf{A} + \mathbf{B}) = \dfrac{d\mathbf{A}}{du} + \dfrac{d\mathbf{B}}{du}$

(ii) $\dfrac{d}{du}(\mathbf{A} \cdot \mathbf{B}) = \mathbf{A} \cdot \dfrac{d\mathbf{B}}{du} + \dfrac{d\mathbf{A}}{du} \cdot \mathbf{B}$

(iii) $\dfrac{d}{du}(\mathbf{A} \times \mathbf{B}) = \mathbf{A} \times \dfrac{d\mathbf{B}}{du} + \dfrac{d\mathbf{A}}{du} \times \mathbf{B}$

(iv) $\dfrac{d}{du}(\phi\mathbf{A}) = \phi\dfrac{d\mathbf{A}}{du} + \dfrac{d\phi}{du}\mathbf{A}$

(v) $\dfrac{d}{du}(\mathbf{A} \cdot \mathbf{B} \times \mathbf{C}) = \mathbf{A} \cdot \mathbf{B} \times \dfrac{d\mathbf{C}}{du} + \mathbf{A} \cdot \dfrac{d\mathbf{B}}{du} \times \mathbf{C} + \dfrac{d\mathbf{A}}{du} \cdot \mathbf{B} \times \mathbf{C}$

(vi) $\dfrac{d}{du}\{\mathbf{A} \times (\mathbf{B} \times \mathbf{C})\} = \mathbf{A} \times \left(\mathbf{B} \times \dfrac{d\mathbf{C}}{du}\right) + \mathbf{A} \times \left(\dfrac{d\mathbf{B}}{du} \times \mathbf{C}\right) + \dfrac{d\mathbf{A}}{du} \times (\mathbf{B} \times \mathbf{C})$

The order of the products in Proposition 2.1 may be important.

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EXAMPLE 3.2 Suppose $\mathbf{A} = 5u^2\mathbf{i} + u\mathbf{j} - u^3\mathbf{k}$ and $\mathbf{B} = \sin u\mathbf{i} - \cos u\mathbf{j}$. Find $\frac{d}{du}(\mathbf{A}\cdot\mathbf{B})$

$$\frac{d}{du}(\mathbf{A}\cdot\mathbf{B}) = \mathbf{A}\cdot\frac{d\mathbf{B}}{du} + \frac{d\mathbf{A}}{du}\cdot\mathbf{B}$$

$$= (5u^2\mathbf{i} + u\mathbf{j} - u^3\mathbf{k})\cdot(\cos u\mathbf{i} + \sin u\mathbf{j}) + (10u\mathbf{i} + \mathbf{j} - 3u^2\mathbf{k})\cdot(\sin u\mathbf{i} - \cos u\mathbf{j})$$

$$= [5u^2\cos u + u\sin u] + [10u\sin u - \cos u]$$

$$= (5u^2 - 1)\cos u + 11u\sin u$$

Another Method

$\mathbf{A}\cdot\mathbf{B} = 5u^2\sin u - u\cos u$. Then

$$\frac{d}{du}(\mathbf{A}\cdot\mathbf{B}) = \frac{d}{du}(5u^2\sin u - ut\cos u) = 5u^2\cos u + 10u\sin u + u\sin u - \cos u$$

$$= (5u^2 - 1)\cos u + 11u\sin u$$

3.4 Partial Derivative of Vectors

Suppose $\mathbf{A}$ is a vector depending on more than one variable, say x, y, z, for example. Then we write $\mathbf{A} = \mathbf{A}(x, y, z)$. The partial derivative of $\mathbf{A}$ with respect to x is denoted and defined as follows when the limit exists:

$$\frac{\partial\mathbf{A}}{\partial x} = \lim_{\Delta x\to 0}\frac{\mathbf{A}(x+\Delta x, y, z) - \mathbf{A}(x, y, z)}{\Delta x}$$

Similarly, the following are the partial derivatives of $\mathbf{A}$ with respect to y and z, respectively, when the limits exist:

$$\frac{\partial\mathbf{A}}{\partial y} = \lim_{\Delta y\to 0}\frac{\mathbf{A}(x, y+\Delta y, z) - \mathbf{A}(x, y, z)}{\Delta y}$$

$$\frac{\partial\mathbf{A}}{\partial z} = \lim_{\Delta z\to 0}\frac{\mathbf{A}(x, y, z+\Delta z) - \mathbf{A}(x, y, z)}{\Delta z}$$

The remarks on continuity and differentiability of functions of one variable can be extended to functions of two or more variables. For example, $\phi(x, y)$ is called continuous at (x, y) if

$$\lim_{\substack{\Delta x\to 0\\ \Delta y\to 0}}\phi(x+\Delta x, y+\Delta y) = \phi(x, y)$$

or if for each positive number ϵ we can find a positive number δ such that

$$|\phi(x+\Delta x, y+\Delta y) - \phi(x, y)| < \epsilon \quad \text{whenever} \quad |\Delta x| < \delta \quad \text{and} \quad |\Delta y| < \delta$$

Similar definitions hold for vector functions of more than two variables.

For functions of two or more variables, we use the term *differentiable* to mean the function has continuous first partial derivatives. (The term is used by others in a slightly weaker sense.)

Higher derivatives can be defined as in calculus. Thus, for example:

$$\frac{\partial^2\mathbf{A}}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial\mathbf{A}}{\partial x}\right), \quad \frac{\partial^2\mathbf{A}}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial\mathbf{A}}{\partial y}\right), \quad \frac{\partial^2\mathbf{A}}{\partial z^2} = \frac{\partial}{\partial z}\left(\frac{\partial\mathbf{A}}{\partial z}\right)$$

$$\frac{\partial^2\mathbf{A}}{\partial x\partial y} = \frac{\partial}{\partial x}\left(\frac{\partial\mathbf{A}}{\partial y}\right), \quad \frac{\partial^2\mathbf{A}}{\partial y\partial x} = \frac{\partial}{\partial y}\left(\frac{\partial\mathbf{A}}{\partial x}\right), \quad \frac{\partial^3\mathbf{A}}{\partial x\partial z^2} = \frac{\partial}{\partial x}\left(\frac{\partial^2\mathbf{A}}{\partial z^2}\right)$$

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In the case that **A** has continuous partial derivatives of the second order at least, we have

$$\frac{\partial^2 \mathbf{A}}{\partial x \partial y} = \frac{\partial^2 \mathbf{A}}{\partial y \partial x}$$

Thus, the order of differentiation does not matter.

EXAMPLE 3.3 Suppose $\phi(x, y, z) = xy^2z$ and $\mathbf{A} = x\mathbf{i} + \mathbf{j} + xy\mathbf{k}$. Find $\dfrac{\partial^3}{\partial x^2 \partial z}(\phi \mathbf{A})$ at the point $P(1, 2, 2)$.

$$\phi \mathbf{A} = x^2y^2z\mathbf{i} + xy^2z\mathbf{j} + x^2y^3z\mathbf{k}$$

$$\frac{\partial}{\partial z}(\phi \mathbf{A}) = x^2y^2\mathbf{i} + xy^2\mathbf{j} + x^2y^3\mathbf{k}$$

$$\frac{\partial^2}{\partial x \partial z}(\phi \mathbf{A}) = 2xy^2\mathbf{i} + y^2\mathbf{j} + 2xy^3\mathbf{k}$$

$$\frac{\partial^3}{\partial x^2 \partial z}(\phi \mathbf{A}) = 2y^2\mathbf{i} + 2y^3\mathbf{k}$$

When $x = 1$, $y = 2$, and $z = 2$, $\dfrac{\partial^3}{\partial x^2 \partial z}(\phi \mathbf{A}) = 8\mathbf{i} + 16\mathbf{k}$.

Rules for partial differentiation of vectors are similar to those in elementary calculus for scalar functions. In particular, the following proposition applies.

PROPOSITION 3.2 Suppose **A** and **B** are vector functions of x, y, z. Then the following laws hold:

(i) $$\frac{\partial}{\partial x}(\mathbf{A} \cdot \mathbf{B}) = \mathbf{A} \cdot \frac{\partial \mathbf{B}}{\partial x} + \frac{\partial \mathbf{A}}{\partial x} \cdot \mathbf{B}$$

(ii) $$\frac{\partial}{\partial x}(\mathbf{A} \times \mathbf{B}) = \mathbf{A} \times \frac{\partial \mathbf{B}}{\partial x} + \frac{\partial \mathbf{A}}{\partial x} \times \mathbf{B}$$

(iii) $$\frac{\partial^2}{\partial y \partial x}(\mathbf{A} \cdot \mathbf{B}) = \frac{\partial}{\partial y}\left\{\frac{\partial}{\partial x}(\mathbf{A} \cdot \mathbf{B})\right\} = \frac{\partial}{\partial y}\left\{\mathbf{A} \cdot \frac{\partial \mathbf{B}}{\partial x} + \frac{\partial \mathbf{A}}{\partial x} \cdot \mathbf{B}\right\}$$

$$= \mathbf{A} \cdot \frac{\partial^2 \mathbf{B}}{\partial y \partial x} + \frac{\partial \mathbf{A}}{\partial y} \cdot \frac{\partial \mathbf{B}}{\partial x} + \frac{\partial \mathbf{A}}{\partial x} \cdot \frac{\partial \mathbf{B}}{\partial y} + \frac{\partial^2 \mathbf{A}}{\partial y \partial x} \cdot \mathbf{B}, \text{ and so on.}$$

The rules for the differentials of vectors are essentially the same as those of elementary calculus as seen in the following proposition.

PROPOSITION 3.3 Suppose **A** and **B** are functions of x, y, z. Then the following laws hold.

(i) If $\mathbf{A} = A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}$, then $d\mathbf{A} = dA_1\mathbf{i} + dA_2\mathbf{j} + dA_3\mathbf{k}$

(ii) $d(\mathbf{A} \cdot \mathbf{B}) = \mathbf{A} \cdot d\mathbf{B} + d\mathbf{A} \cdot \mathbf{B}$

(iii) $d(\mathbf{A} \times \mathbf{B}) = \mathbf{A} \times d\mathbf{B} + d\mathbf{A} \times \mathbf{B}$

(iv) If $\mathbf{A} = \mathbf{A}(x, y, z)$, then $d\mathbf{A} = \dfrac{\partial \mathbf{A}}{\partial x}dx + \dfrac{\partial \mathbf{A}}{\partial y}dy + \dfrac{\partial \mathbf{A}}{\partial z}dz$, and so on.

3.5 Differential Geometry

Differential geometry involves the study of curves and surfaces. Suppose C is a space curve defined by the function $\mathbf{r}(u)$. Then, we have seen that $d\mathbf{r}/du$ is a vector in the direction of the tangent to C. Suppose the scalar u is taken as the arc length s measured from some fixed point on C. Then $d\mathbf{r}/ds$ is a unit tangent vector to C and it is denoted by $\mathbf{T}$ (see Fig. 3-3). The rate at which $\mathbf{T}$ changes with respect to s is a measure of the curvature of C and is given by $d\mathbf{T}/ds$. The direction of $d\mathbf{T}/ds$ at any given point on C is normal to the curve at that point (see Problem 3.9). If $\mathbf{N}$ is a unit vector in this normal direction, it is called the *principal normal*

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to the curve. Then $d\mathbf{T}/ds = \kappa\mathbf{N}$, where κ is called the *curvature* of C at the specified point. The quantity $\rho = 1/\kappa$ is called the *radius of curvature*.

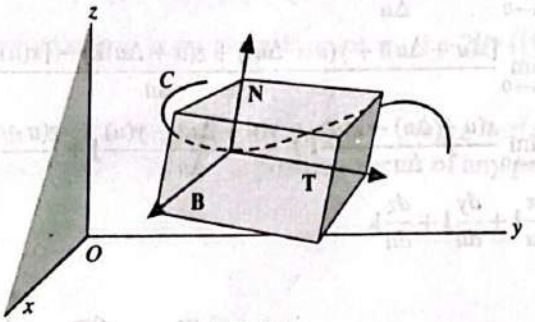


Fig. 3-3

A unit vector $\mathbf{B}$ perpendicular to the plane of $\mathbf{T}$ and $\mathbf{N}$ and such that $\mathbf{B} = \mathbf{T} \times \mathbf{N}$, is called the *binormal* to the curve. It follows that directions $\mathbf{T}$, $\mathbf{N}$, $\mathbf{B}$ form a localized right-handed rectangular coordinate system at any specified point of C. This coordinate system is called the *trihedral* or *triad* at the point. As s changes, the coordinate system moves and is known as the *moving trihedral*.

Frenet–Serret Formulas

A set of relations involving derivatives of the fundamental vectors $\mathbf{T}$, $\mathbf{N}$, and $\mathbf{B}$ is known collectively as the *Frenet–Serret formulas* given by

$$\frac{d\mathbf{T}}{ds} = \kappa\mathbf{N}, \quad \frac{d\mathbf{N}}{ds} = \tau\mathbf{B} - \kappa\mathbf{T}, \quad \frac{d\mathbf{B}}{ds} = -\tau\mathbf{N}$$

where τ is a scalar called the *torsion*. The quantity $\sigma = 1/\tau$ is called the *radius of torsion*.

The *osculating plane* to a curve at a point P is the plane containing the tangent and principal normal at P. The *normal plane* is the plane through P perpendicular to the tangent. The *rectifying plane* is the plane through P, which is perpendicular to the principal normal.

Mechanics

Mechanics often includes the study of the motion of particles along curves. (This study being known as *kinematics*.) In this area, some of the results of differential geometry can be of value.

A study of forces on moving objects is considered in *dynamics*. Fundamental to this study is Newton's famous law which states that if $\mathbf{F}$ is the net force acting on an object of mass m moving with velocity $\mathbf{v}$, then

$$\mathbf{F} = \frac{d}{dt}(m\mathbf{v})$$

where $m\mathbf{v}$ is the momentum of the object. If m is constant, this becomes $\mathbf{F} = m(d\mathbf{v}/dt) = m\mathbf{a}$, where $\mathbf{a}$ is the acceleration of the object.

SOLVED PROBLEMS

3.1. Suppose $\mathbf{R}(u) = x(u)\mathbf{i} + y(u)\mathbf{j} + z(u)\mathbf{k}$, where x, y, and z are differentiable functions of a scalar u. Prove that

$$\frac{d\mathbf{R}}{du} = \frac{dx}{du}\mathbf{i} + \frac{dy}{du}\mathbf{j} + \frac{dz}{du}\mathbf{k}.$$

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Solution

$$\frac{d\mathbf{R}}{du} = \lim_{\Delta u \to 0} \frac{\mathbf{R}(u+\Delta u) - \mathbf{R}(u)}{\Delta u}$$

$$= \lim_{\Delta u \to 0} \frac{[x(u+\Delta u)\mathbf{i} + y(u+\Delta u)\mathbf{j} + z(u+\Delta u)\mathbf{k}] - [x(u)\mathbf{i} + y(u)\mathbf{j} + z(u)\mathbf{k}]}{\Delta u}$$

$$= \lim_{\Delta u \to 0} \frac{x(u+\Delta u) - x(u)}{\Delta u}\mathbf{i} + \frac{y(u+\Delta u) - y(u)}{\Delta u}\mathbf{j} + \frac{z(u+\Delta u) - z(u)}{\Delta u}\mathbf{k}$$

$$= \frac{dx}{du}\mathbf{i} + \frac{dy}{du}\mathbf{j} + \frac{dz}{du}\mathbf{k}$$

3.2. Given $\mathbf{R} = (3\cos t)\mathbf{i} + (3\sin t)\mathbf{j} + (4t)\mathbf{k}$. Find: (a) $\dfrac{d\mathbf{R}}{dt}$, (b) $\dfrac{d^2\mathbf{R}}{dt}$, (c) $\left|\dfrac{d\mathbf{R}}{dt}\right|$, (d) $\left|\dfrac{d^2\mathbf{R}}{dt}\right|$.

Solution

(a) $\dfrac{d\mathbf{R}}{dt} = \dfrac{d}{dt}(3\cos t)\mathbf{i} + \dfrac{d}{dt}(3\sin t)\mathbf{j} + \dfrac{d}{dt}(4t)\mathbf{k} = (-3\sin t)\mathbf{i} + (3\cos t)\mathbf{j} + 4\mathbf{k}.$

(b) $\dfrac{d^2\mathbf{R}}{dt} = \dfrac{d}{dt}\left(\dfrac{d\mathbf{R}}{dt}\right) = \dfrac{d}{dt}(-3\sin t)\mathbf{i} + \dfrac{d}{dt}(3\cos t)\mathbf{j} + \dfrac{d}{dt}(4)\mathbf{k} = (-3\cos t)\mathbf{i} + (-3\sin t)\mathbf{j}.$

(c) $\left|\dfrac{d\mathbf{R}}{dt}\right| = [(-3\sin t)^2 + (3\cos t)^2 + (4)^2]^{1/2} = 5.$

(d) $\left|\dfrac{d^2\mathbf{R}}{dt}\right| = [(-3\cos t)^2 + (-3\sin t)^2 + (0)^2]^{1/2} = 3.$

3.3. A particle moves along a curve whose parametric equations are $x = e^{-t}$, $y = 2\cos 3t$, $z = 2\sin 3t$, where t is the time.

(a) Determine its velocity and acceleration at any time.

(b) Find the magnitudes of the velocity and acceleration at $t = 0$.

Solution

(a) The position vector $\mathbf{r}$ of the particle is $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k} = e^{-t}\mathbf{i} + 2\cos 3t\mathbf{j} + 2\sin 3t\mathbf{k}$. Then the velocity is $\mathbf{v} = d\mathbf{r}/dt = -e^{-t}\mathbf{i} - 6\sin 3t\mathbf{j} + 6\cos 3t\mathbf{k}$ and the acceleration is $\mathbf{a} = d^2\mathbf{r}/dt^2 = e^{-t}\mathbf{i} - 18\cos 3t\mathbf{j} - 18\sin 3t\mathbf{k}$.

(b) At $t = 0$, $d\mathbf{r}/dt = -\mathbf{i} + 6\mathbf{k}$ and $d^2\mathbf{r}/dt^2 = \mathbf{i} - 18\mathbf{j}$. Then

magnitude of velocity at $t = 0$ is $\sqrt{(-1)^2 + (6)^2} = \sqrt{37}$

magnitude of acceleration at $t = 0$ is $\sqrt{(1)^2 + (-18)^2} = \sqrt{325}$.

3.4. A particle moves along the curve $x = 2t^2$, $y = t^2 - 4t$, $z = -t - 5$ where t is the time. Find the components of its velocity and acceleration at time $t = 1$ in the direction $\mathbf{i} - 2\mathbf{j} + 2\mathbf{k}$.

Solution

$$\text{Velocity} = \frac{d\mathbf{r}}{dt} = \frac{d}{dt}[(2t^2)\mathbf{i} + (t^2 - 4t)\mathbf{j} + (-t - 5)\mathbf{k}]$$

$$= (4t)\mathbf{i} + (2t - 4)\mathbf{j} - \mathbf{k} = 4\mathbf{i} - 2\mathbf{j} - \mathbf{k} \quad \text{at } t = 1.$$

Unit vector in direction $\mathbf{i} - 2\mathbf{j} + 2\mathbf{k}$ is $\dfrac{\mathbf{i} - 2\mathbf{j} + 2\mathbf{k}}{\sqrt{(1)^2 + (-2)^2 + (2)^2}} = \frac{1}{3}\mathbf{i} - \frac{2}{3}\mathbf{j} + \frac{2}{3}\mathbf{k}$.

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Then the component in the given direction is $(4\mathbf{i} - 2\mathbf{j} - \mathbf{k}) \cdot (\frac{1}{3}\mathbf{i} - \frac{2}{3}\mathbf{j} + \frac{2}{3}\mathbf{k}) = 2$

$$\text{Acceleration} = \frac{d^2\mathbf{r}}{dt^2} = \frac{d}{dt}\left(\frac{d\mathbf{r}}{dt}\right) = \frac{d}{dt}[(4t)\mathbf{i} + (2t - 4)\mathbf{j} - \mathbf{k}] = 4\mathbf{i} + 2\mathbf{j}.$$

Then the component of the acceleration in the given direction is $(4\mathbf{i} + 2\mathbf{j}) \cdot (\frac{1}{3}\mathbf{i} - \frac{2}{3}\mathbf{j} + \frac{2}{3}\mathbf{k}) = 0$.

3.5. A curve C is defined by parametric equations $x = x(s)$, $y = y(s)$, $z = z(s)$, where s is the arc length of C measured from a fixed point on C. If $\mathbf{r}$ is the position vector of any point on C, show that $d\mathbf{r}/ds$ is a unit vector tangent to C.

Solution

The vector

$$\frac{d\mathbf{r}}{ds} = \frac{d}{ds}(x\mathbf{i} + y\mathbf{j} + z\mathbf{k}) = \frac{dx}{ds}\mathbf{i} + \frac{dy}{ds}\mathbf{j} + \frac{dz}{ds}\mathbf{k}$$

is tangent to the curve $x = x(s)$, $y = y(s)$, $z = z(s)$. To show that it has unit magnitude, we note that

$$\left|\frac{d\mathbf{r}}{ds}\right| = \sqrt{\left(\frac{dx}{ds}\right)^2 + \left(\frac{dy}{ds}\right)^2 + \left(\frac{dz}{ds}\right)^2} = \sqrt{\frac{(dx)^2 + (dy)^2 + (dz)^2}{(ds)^2}} = 1$$

since $(ds)^2 = (dx)^2 + (dy)^2 + (dz)^2$ from the calculus.

3.6. (a) Find the unit tangent vector to any point on the curve $x = t^2 - t$, $y = 4t - 3$, $z = 2t^2 - 8t$.

(b) Determine the unit tangent at the point where $t = 2$.

Solution

(a) A tangent to the curve at any point is

$$\frac{d\mathbf{r}}{dt} = \frac{d}{dt}[(t^2 - t)\mathbf{i} + (4t - 3)\mathbf{j} + (2t^2 - 8t)\mathbf{k}] = (2t - 1)\mathbf{i} + 4\mathbf{j} + (4t - 8)\mathbf{k}.$$

The magnitude of the vector is $|d\mathbf{r}/dt| = [(2t - 1)^2 + (4)^2 + (4t - 8)^2]^{1/2}$. Then the required unit tangent vector is

$$\mathbf{T} = [(2t - 1)\mathbf{i} + 4\mathbf{j} + (4t - 8)\mathbf{k}]/[(2t - 1)^2 + (4)^2 + (4t - 8)^2]^{1/2}.$$

Note that since $|d\mathbf{r}/dt| = ds/dt$, we have

$$\mathbf{T} = \frac{d\mathbf{r}/dt}{ds/dt} = \frac{d\mathbf{r}}{ds}.$$

(b) At $t = 2$, the unit tangent vector is $\mathbf{T} = \dfrac{3\mathbf{i} + 4\mathbf{j}}{\sqrt{(3)^2 + (4)^2 + 0^2}} = \frac{3}{5}\mathbf{i} + \frac{4}{5}\mathbf{j}$.

3.7. Suppose $\mathbf{A}$ and $\mathbf{B}$ are differentiable functions of a scalar u. Prove:

(a) $\dfrac{d}{du}(\mathbf{A} \cdot \mathbf{B}) = \mathbf{A} \cdot \dfrac{d\mathbf{B}}{du} + \dfrac{d\mathbf{A}}{du} \cdot \mathbf{B}$, (b) $\dfrac{d}{du}(\mathbf{A} \times \mathbf{B}) = \mathbf{A} \times \dfrac{d\mathbf{B}}{du} + \dfrac{d\mathbf{A}}{du} \times \mathbf{B}$

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3.13 Let $\mathbf{A} = (2x^2y - x^4)\mathbf{i} + (e^{xy} - y\sin x)\mathbf{j} + (x^2\cos y)\mathbf{k}$. Find: (a) $\dfrac{\partial \mathbf{A}}{\partial x}$, (b) $\dfrac{\partial \mathbf{A}}{\partial y}$.

Solution

(a) $$\frac{\partial \mathbf{A}}{\partial x} = \frac{\partial}{\partial x}(2x^2y - x^4)\mathbf{i} + \frac{\partial}{\partial x}(e^{xy} - y\sin x)\mathbf{j} + \frac{\partial}{\partial x}(x^2\cos y)\mathbf{k}$$
$$= (4xy - 4x^3)\mathbf{i} + (ye^{xy} - y\cos x)\mathbf{j} + 2x\cos y\mathbf{k}$$

(b) $$\frac{\partial \mathbf{A}}{\partial y} = \frac{\partial}{\partial y}(2x^2y - x^4)\mathbf{i} + \frac{\partial}{\partial y}(e^{xy} - y\sin x)\mathbf{j} + \frac{\partial}{\partial y}(x^2\cos y)\mathbf{k}$$
$$= 2x^2\mathbf{i} + (xe^{xy} - \sin x)\mathbf{j} - x^2\sin y\mathbf{k}$$

3.14. Let $\mathbf{A}$ be the vector in Problem 3.13. Find: (a) $\dfrac{\partial^2 \mathbf{A}}{\partial x^2}$, (b) $\dfrac{\partial^2 \mathbf{A}}{\partial y^2}$.

Solution

(a) $$\frac{\partial^2 \mathbf{A}}{\partial x^2} = \frac{\partial}{\partial x}(4xy - 4x^3)\mathbf{i} + \frac{\partial}{\partial x}(ye^{xy} - y\cos x)\mathbf{j} + \frac{\partial}{\partial x}(2x\cos y)\mathbf{k}$$
$$= (4y - 12x^2)\mathbf{i} + (y^2e^{xy} + y\sin x)\mathbf{j} + 2\cos y\mathbf{k}$$

(b) $$\frac{\partial^2 \mathbf{A}}{\partial y^2} = \frac{\partial}{\partial y}(2x^2)\mathbf{i} + \frac{\partial}{\partial y}(xe^{xy} - \sin x)\mathbf{j} - \frac{\partial}{\partial y}(x^2\sin y)\mathbf{k}$$
$$= 0 + x^2e^{xy}\mathbf{j} - x^2\cos y\mathbf{k} = x^2e^{xy}\mathbf{j} - x^2\cos y\mathbf{k}$$

3.15. Let $\mathbf{A}$ be the vector in Problem 3.13. Find: (a) $\dfrac{\partial^2 \mathbf{A}}{\partial x \partial y}$, (b) $\dfrac{\partial^2 \mathbf{A}}{\partial y \partial x}$.

Solution

(a) $$\frac{\partial^2 \mathbf{A}}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial \mathbf{A}}{\partial y}\right) = \frac{\partial}{\partial x}(2x^2)\mathbf{i} + \frac{\partial}{\partial x}(xe^{xy} - \sin x)\mathbf{j} - \frac{\partial}{\partial x}(x^2\sin y)\mathbf{k}$$
$$= 4x\mathbf{i} + (xye^{xy} + e^{xy} - \cos x)\mathbf{j} - 2x\sin y\mathbf{k}$$

(b) $$\frac{\partial^2 \mathbf{A}}{\partial y \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial \mathbf{A}}{\partial x}\right) = \frac{\partial}{\partial y}(4xy - 4x^3)\mathbf{i} + \frac{\partial}{\partial y}(ye^{xy} - y\cos x)\mathbf{j} + \frac{\partial}{\partial y}(2x\cos y)\mathbf{k}$$
$$= 4x\mathbf{i} + (xye^{xy} + e^{xy} - \cos x)\mathbf{j} - 2x\sin y\mathbf{k}$$

Note that $\partial^2\mathbf{A}/\partial y\partial x = \partial^2\mathbf{A}/\partial x\partial y$, that is, the order of differentiation is immaterial. This is true in general if $\mathbf{A}$ has continuous partial derivatives of the second order at least.

3.16. Suppose $\phi(x, y, z) = xy^2z$ and $\mathbf{A} = xz\mathbf{i} - xy^2\mathbf{j} + yz^2\mathbf{k}$. Find $\dfrac{\partial^3}{\partial x^2 \partial z}(\phi\mathbf{A})$ at the point $(2, -1, 1)$.

Solution

$$\phi\mathbf{A} = (xy^2z)(xz\mathbf{i} - xy^2\mathbf{j} + yz^2\mathbf{k}) = x^2y^2z^2\mathbf{i} - x^2y^4z\mathbf{j} + xy^3z^3\mathbf{k}$$

$$\frac{\partial}{\partial z}(\phi\mathbf{A}) = \frac{\partial}{\partial z}(x^2y^2z^2\mathbf{i} - x^2y^4z\mathbf{j} + xy^3z^3\mathbf{k}) = 2x^2y^2z\mathbf{i} - x^2y^4\mathbf{j} + 3xy^3z^2\mathbf{k}$$

$$\frac{\partial^2}{\partial x \partial z}(\phi\mathbf{A}) = \frac{\partial}{\partial x}(2x^2y^2z\mathbf{i} - x^2y^4\mathbf{j} + 3xy^3z^2\mathbf{k}) = 4xy^2z\mathbf{i} - 2xy^4\mathbf{j} + 3y^3z^2\mathbf{k}$$

$$\frac{\partial^3}{\partial x^2 \partial z}(\phi\mathbf{A}) = \frac{\partial}{\partial x}(4xy^2z\mathbf{i} - 2xy^4\mathbf{j} + 3y^3z^2\mathbf{k}) = 4y^2z\mathbf{i} - 2y^4\mathbf{j}$$

If $x = 2$, $y = -1$, and $z = 1$, this becomes $4(-1)^2(1)\mathbf{i} - 2(-1)^4\mathbf{j} = 4\mathbf{i} - 2\mathbf{j}$.

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Then $\frac{d\mathbf{i}}{dt} = \alpha_1\mathbf{j} + \alpha_2\mathbf{k}$, $\frac{d\mathbf{j}}{dt} = \alpha_3\mathbf{k} - \alpha_1\mathbf{i}$, $\frac{d\mathbf{k}}{dt} = -\alpha_2\mathbf{i} - \alpha_3\mathbf{j}$ and

$$A_1\frac{d\mathbf{i}}{dt} + A_2\frac{d\mathbf{j}}{dt} + A_3\frac{d\mathbf{k}}{dt} = (-\alpha_1A_2 - \alpha_2A_3)\mathbf{i} + (\alpha_1A_1 - \alpha_3A_3)\mathbf{j} + (\alpha_2A_1 + \alpha_3A_2)\mathbf{k}$$ which can be written as

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \alpha_3 & -\alpha_2 & \alpha_1 \\ A_1 & A_2 & A_3 \end{vmatrix}$$

Then if we choose $\alpha_3 = \omega_1, -\alpha_2 = \omega_2, \alpha_1 = \omega_3$ the determinant becomes

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \omega_1 & \omega_2 & \omega_3 \\ A_1 & A_2 & A_3 \end{vmatrix} = \boldsymbol{\omega} \times \mathbf{A}$$

where $\boldsymbol{\omega} = \omega_1\mathbf{i} + \omega_2\mathbf{j} + \omega_3\mathbf{k}$. The quantity $\boldsymbol{\omega}$ is the angular velocity vector of the moving system with respect to the fixed system.

(b) By definition $D_f\mathbf{A} = \left.\frac{d\mathbf{A}}{dt}\right|_f$ = derivative in fixed system

$D_m\mathbf{A} = \left.\frac{d\mathbf{A}}{dt}\right|_m$ = derivative in moving system.

From (a),

$$D_f\mathbf{A} = D_m\mathbf{A} + \boldsymbol{\omega} \times \mathbf{A} = (D_m + \boldsymbol{\omega} \times)\mathbf{A}$$

and shows the equivalence of the operators $D_f \equiv D_m + \boldsymbol{\omega}\times$.

SUPPLEMENTARY PROBLEMS

3.31. Suppose $\mathbf{R} = e^{-t}\mathbf{i} + \ln(t^2+1)\mathbf{j} - \tan t\mathbf{k}$. Find: (a) $d\mathbf{R}/dt$, (b) $d^2\mathbf{R}/dt^2$, (c) $|d\mathbf{R}/dt|$, (d) $|d^2\mathbf{R}/dt^2|$ at $t = 0$.

3.32. Suppose a particle moves along the curve $x = 2\sin 3t, y = 2\cos 3t, z = 8t$ at any time $t > 0$.
(a) Find the velocity and acceleration of the particle.
(b) Find the magnitude of the velocity and acceleration.

3.33. Find a unit tangent vector to any point on the curve $x = a\cos\omega t$, $y = a\sin\omega t$, $z = bt$ where a, b, and ω are constants.

3.34 Suppose $\mathbf{A} = t^2\mathbf{i} - t\mathbf{j} + (2t+1)\mathbf{k}$ and $\mathbf{B} = (2t-3)\mathbf{i} + \mathbf{j} - t\mathbf{k}$. Find
(a) $\frac{d}{dt}(\mathbf{A}\cdot\mathbf{B})$, (b) $\frac{d}{dt}(\mathbf{A}\times\mathbf{B})$, (c) $\frac{d}{dt}|\mathbf{A}+\mathbf{B}|$, (d) $\frac{d}{dt}\left(\mathbf{A}\times\frac{d\mathbf{B}}{du}\right)$ at $t = 1$.

3.35. Suppose $\mathbf{A} = \sin u\mathbf{i} + \cos u\mathbf{j} + u\mathbf{k}$, $\mathbf{B} = \cos u\mathbf{i} - \sin u\mathbf{j} - 3\mathbf{k}$, and $\mathbf{C} = 2\mathbf{i} + 3\mathbf{j} - \mathbf{k}$.
Find $\frac{d}{du}(\mathbf{A}\times(\mathbf{B}\times\mathbf{C}))$ at $u = 0$.

3.36. Show: (a) $\frac{d}{ds}\left(\mathbf{A}\cdot\frac{d\mathbf{B}}{ds} - \frac{d\mathbf{A}}{ds}\cdot\mathbf{B}\right) = \mathbf{A}\frac{d^2\mathbf{B}}{ds^2} - \frac{d^2\mathbf{A}}{ds^2}\cdot\mathbf{B}$ where $\mathbf{A}$ and $\mathbf{B}$ are differential functions of s.

(b) $\frac{d}{ds}\left(\mathbf{A}\times\frac{d\mathbf{B}}{ds} - \frac{d\mathbf{A}}{ds}\times\mathbf{B}\right) = \mathbf{A}\times\frac{d^2\mathbf{B}}{ds^2} - \frac{d^2\mathbf{A}}{ds^2}\times\mathbf{B}$

(c) $\frac{d}{dt}\left(\mathbf{V}\cdot\frac{d\mathbf{V}}{dt}\times\frac{d^2\mathbf{V}}{dt^2}\right) = \mathbf{V}\cdot\frac{d\mathbf{V}}{dt}\times\frac{d^3\mathbf{V}}{dt^3}$

3.37. Suppose $\mathbf{A}(t) = 3t^2\mathbf{i} - (t+4)\mathbf{j} + (t^2-2t)\mathbf{k}$ and $\mathbf{B}(t) = \sin t\mathbf{i} + 3e^{-t}\mathbf{j} - 3\cos t\mathbf{k}$. Find $\frac{d^2}{dt^2}(\mathbf{A}\times\mathbf{B})$ at $t = 0$.

3.38. Let $\frac{d^2\mathbf{A}}{dt^2} = 6t\mathbf{i} - 24t^2\mathbf{j} + 4\sin t\mathbf{k}$. Find $\mathbf{A}$ given that $\mathbf{A} = 2\mathbf{i} + \mathbf{j}$ and $\frac{d\mathbf{A}}{dt} = -\mathbf{i} - 3\mathbf{k}$ at $t = 0$.

3.39. Show that $\mathbf{r} = e^{-t}(\mathbf{C}_1\cos 2t + \mathbf{C}_2\sin 2t)$, where $\mathbf{C}_1$ and $\mathbf{C}_2$ are constant vectors, is a solution of the differential equation $\frac{d^2\mathbf{r}}{dt^2} + 2\frac{d\mathbf{r}}{dt} + 5\mathbf{r} = 0$.

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